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Inside Forced Liquidation: Margin Constraints, Execution, and Risk Transfer

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ABSTRACT

Forced liquidation is commonly treated as the endpoint of financial distress. Yet, liquidation is a resolution process: margin losses create inventory, execution rules dictate position releases, and recipient balance sheets absorb the risk. We link account-wide margins to liquidations, executions, and recipients in Hyperliquid. Using regression discontinuity and shift-share designs, we identify the loss-to-liquidation and liquidation-to-price mappings. An event reconstruction traces downstream risk transfer during the October 2025 cascade. Fragility is concentrated near maintenance boundaries. Less aggressive execution reduces trading costs, price pressure, and subsequent liquidations. Backstops can themselves later face deleveraging. Liquidation reallocates risk rather than eliminating it.

Keywords: forced liquidation; margin constraints; liquidation policy; price pressure; risk transfer; perpetual futures

JEL Classification: G01; G12; G14; G23

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1 Introduction

Forced liquidation is commonly treated as the endpoint of financial distress: an account breaches a constraint, its position is closed, and the resulting liquidation volume measures the severity of the event. Economically, however, liquidation is not an endpoint but a resolution process. Closing a distressed account does not eliminate its position. The inventory must be executed in the market, transferred to a backstop, or reassigned through a loss-sharing mechanism. The market consequences of a margin shock therefore depend not only on how much leverage exists, but also on which accounts are close to their maintenance boundaries, how the platform releases their positions, and whether the receiving balance sheets can retain the risk.

Existing research establishes important parts of this mechanism but rarely observes the entire sequence. Theories of collateral amplification and funding liquidity connect losses, binding constraints, forced sales, and deteriorating liquidity (Kiyotaki and Moore, 1997; Brunnermeier and Pedersen, 2009; Geanakoplos, 2010; Acharya and Viswanathan, 2011; Gârleanu and Pedersen, 2011). Empirical studies show that constrained investors sell and that distressed sales can move prices (Coval and Stafford, 2007; Ellul et al., 2011). Yet, the relevant objects are generally observed in separate datasets: portfolio and margin states without protocol-labeled forced orders, forced transactions without the balance-sheet losses that produced them, or executions without the identities and subsequent condition of the recipients. As a result, forced liquidation is often measured as a quantity of distressed trading rather than as a sequence of balance-sheet transfers governed by an explicit resolution rule.

This separation leaves three central questions unanswered. First, what determines how much forced inventory a given price shock creates? Aggregate leverage is insufficient if liquidation depends on the distribution of accounts near their maintenance boundaries. Second, conditional on an account entering liquidation, does the rule governing how quickly its inventory reaches the market independently affect execution costs, local prices, and the liquidation of other accounts? Liquidation volume alone cannot answer this question because the same position can be released immediately or gradually. Third, where does the risk go after liquidation? A backstop may stabilize the distressed account while concentrating exposure on another balance sheet that can itself later become constrained.

These questions are also necessary for distinguishing two interpretations of liquidation

cascades. Under a passive-liquidation view, an external shock simultaneously lowers prices and triggers liquidations; liquidation volume records the shock’s consequences but does not independently worsen them. Under an amplification view, the shock pushes fragile accounts across maintenance boundaries, creating price-insensitive inventory whose execution depresses local prices and compresses other accounts’ margin buffers. Aggregate comovement between prices and liquidations cannot distinguish these views. Doing so requires separately identifying the mapping from portfolio losses to forced inventory, the mapping from forced inventory to local price pressure, and the mapping from execution pressure to subsequent liquidation. It also requires following the inventory to the balance sheets that receive it.

We study this sequence in Hyperliquid, a transparent, fully on-chain market for perpetual futures. The empirical advantage is not granularity but closure. We link account-wide portfolios and maintenance states to protocol-labeled liquidations, every execution, and the balance sheet that first receives each position. The setting also distinguishes the official spot-oracle price that measures the external asset-price shock, the mark price that determines liquidation eligibility, and transaction prices that record local execution. Cross margin makes the relevant shock account-wide: an asset can be liquidated because losses elsewhere consume the same collateral pool. The broad panel follows 190,855 accounts across 195 assets and 215 windows, yielding 29.5 million account–asset–window observations; the event replay reconstructs the cascade account by account, minute by minute, and fill by fill.

Our empirical architecture mirrors the economic sequence and separates link-level identification from system-level accounting. First, a shock-exposure design combines predetermined account portfolios with subsequent asset-specific oracle innovations to identify how external losses move fragile balance sheets into liquidation and create forced inventory at entry. Second, among positions already entering market-method liquidation near the cutoff, a discontinuity in the platform’s liquidation rule changes the initial quantity submitted to the market, allowing us to identify how execution aggressiveness affects trading costs, immediate local prices, and the subsequent liquidation of other accounts in the same asset. Finally, the October 10 reconstruction provides high-resolution event accounting for how these three identified links appear in a system-wide cascade, measuring valuation-path exposure, initial recipients, and downstream ADL. The shock-exposure design identifies the production link; the liquidation-rule discontinuity identifies the execution link and its feedback into subsequent liquidation; and the event reconstruction accounts for the system-level quantities and

balance-sheet transfers observed alongside them.

The evidence from these analyses yields three main findings. First, the production of forced inventory depends on the distribution of margin buffers across accounts, not aggregate leverage alone. Across the 215-window panel, discretionary position reduction is widespread under moderate margin pressure, whereas forced liquidation is sharply concentrated among accounts with the thinnest margin buffers. During October 10, accounts in similar predetermined risk states experience different losses because their frozen portfolios load differently on contemporaneous asset-specific oracle innovations. A loss equal to 10% of frozen equity raises the next-minute probability of first liquidation by 2.02 percentage points, relative to an unconditional probability of 0.49%, and raises the entry-minute forced-inventory rate by 1.72 percentage points. The quantity of forced inventory generated by a common shock therefore depends on the cross-sectional distribution of margin buffers, not simply on aggregate leverage or the size of the market decline.

Second, liquidation policy also functions as liquidity policy. By governing the initial release of forced inventory, it affects execution costs and local price pressure, with additional evidence of consequences for subsequent liquidation. We establish these effects by comparing otherwise similar liquidated positions near the platform’s \$100,000 policy threshold, where the liquidation rule sharply changes the initial share submitted to the market. Assignment to the less aggressive regime lowers the initial liquidation-order share by 50.0 percentage points, adverse execution slippage by 15.6 basis points, and the maximum adverse transaction-price movement during the first ten seconds by 15.4 basis points. The price effects are largest in stressed market conditions. The same discontinuity provides additional evidence that less aggressive execution reduces the number of other accounts in the same asset entering liquidation over the following five minutes. Execution policy therefore governs not only the cost of resolving one account but potentially how far that resolution propagates.

The October 10 reconstruction next provides high-resolution accounting of valuation-path exposure and subsequent forced flow. During the liquidation burst, forced inventory arrives as perpetual-futures prices fall relative to the spot-oracle benchmark, trading costs rise, and market depth contracts. Relative to the official spot-oracle benchmark, the realized mark path is associated with \$161.9 million more gross position notional in margin shortfall. Accounts classified only along that path subsequently account for \$242.6 million of forced liquidation.

Our third finding is that a backstop is part of the system’s balance-sheet network rather than a terminal destination. Accounts outside the identified Hyperliquidity Provider (HLP) and system-account set generate \$4.667 billion of initial forced liquidation, nearly half of which is initially received by identified HLP/system accounts. Accounts in the same identified set later appear on the forced side of \$1.999 billion in automatic deleveraging (ADL)-linked trades.¹ Initial receipt by the backstop therefore need not be the final observed allocation in the forced-transfer sequence.

Taken together, the evidence shows the limits of three common shortcuts in measuring financial fragility. Aggregate leverage does not summarize how much forced inventory a shock will create; forced-sale volume does not determine price impact independently of the execution rule; and the presence of a backstop does not imply that risk has left the system. The consequences of forced liquidation are therefore governed by a sequence of balance-sheet mappings, not simply by the initial price decline or the volume liquidated.

These findings contribute to the literature in three ways. First, we introduce an empirical decomposition of forced liquidation into three distinct mappings: the production of forced inventory, its release into the market, and its transfer across recipient balance sheets. This approach extends the closest account-level and event-reconstruction studies by shifting attention from trading responses to the institutional resolution of binding constraints. Bian et al. (2026) show that proximity to margin calls affects account trading and transmits shocks across assets. We examine the stage that begins once discretion ends: protocol-generated orders, the rule determining their initial release, and the resulting execution prices and recipients. Kirilenko et al. (2017) reconstruct account inventories during the Flash Crash; we connect a comparable transaction-level audit trail to observed maintenance states and quasi-experimental variation in both portfolio losses and liquidation policy. The resulting framework distinguishes whether a cascade reflects the creation of more forced inventory, the more aggressive execution of a given inventory stock, or the inability of recipient balance sheets to retain the transferred risk.

Second, we identify liquidation policy as an independent determinant of execution costs and local price pressure, and provide evidence on subsequent liquidation risk. Empirical

¹We identify HLP/system accounts using a versioned registry that includes the community-funded protocol liquidator vault. ADL involuntarily reduces selected opposing positions; Internet Appendix A documents recipient matching and flow classification.

studies typically treat forced sales as a quantity of non-discretionary order flow (Shleifer, 1986; Coval and Stafford, 2007; Ellul et al., 2011; Falato et al., 2021). Around the policy threshold, the comparison is among positions already entering market-method liquidation, while the platform changes the initial share submitted to the book. The discontinuity therefore separates the production of forced inventory from its release into the market. Reducing the initial share materially lowers the distressed trader’s execution costs and the immediate local price dislocation, and is associated with fewer other accounts subsequently liquidated. A balance-sheet failure has no unique market effect independent of the protocol used to resolve it.

Third, we make recipient balance sheets an observable part of forced-liquidation resolution. Public-market counterparties, the platform backstop, and other identified system accounts initially receive the inventory; accounts in the same identified institutional set later reappear on the forced side of the system’s loss-sharing mechanism. This evidence connects the forced-transfer accounting to theories of limited intermediary capital and cash-in-the-market resolution, in which the allocation of distressed assets depends on the financing capacity of potential acquirers (Shleifer and Vishny, 1997; Gromb and Vayanos, 2002; Acharya and Yorulmazer, 2008; Duffie, 2010; He and Krishnamurthy, 2013). The relevant empirical object is therefore not only how much inventory is liquidated, but also which balance sheet first receives it and how the same institutional set enters downstream forced reallocation.

Digital-asset markets make normally latent crisis mechanisms observable. Liu et al. (2023) study strategic withdrawals during a run, Lehar and Parlour (2022) study collateral-liquidation feedback, and Chitra (2025) and Chitra et al. (2026) study ADL design and performance in the same event. We follow the account-to-market sequence from predetermined portfolio losses through forced execution to downstream risk transfer.

The remainder of the paper proceeds as follows. Section 2 describes Hyperliquid’s margin system, liquidation rules, and risk-transfer mechanisms and develops the conceptual framework. Section 3 introduces the linked data and documents the distribution of margin fragility and private portfolio adjustment. Section 4 identifies how losses on predetermined portfolios generate forced inventory. Section 5 uses the liquidation-policy discontinuity to estimate the effects of execution aggressiveness on trading costs, local prices, and subsequent liquidations. Section 6 reconstructs the October 10 cascade, tracing valuation-path exposure, initial recipients, and downstream automatic deleveraging. Section 7 concludes.

2 Institutional Setting and Conceptual Framework

This section describes the institutional features that turn balance-sheet losses into forced liquidation and govern how forced positions reach the market and move across recipient balance sheets. Three features are central. First, a maintenance boundary separates discretionary adjustment from mechanically executed liquidation. Second, the platform distinguishes the underlying spot oracle, the mark used for account valuation and risk management, and the transaction prices at which liquidation orders execute. Third, liquidation proceeds through ordinary market execution, a protocol backstop that takes over distressed portfolios when book execution cannot restore compliance, and, when necessary, automatic deleveraging. Because maintenance is evaluated at the account level for cross-margin positions, the relevant shock exposure is the loss on the account’s predetermined portfolio. Together, these features organize the empirical analysis around the production of forced inventory, its local execution, and downstream risk transfer. Figure 1 summarizes this sequence and distinguishes the three identified links from remaining event-path propagation and risk-transfer accounting. The third link is identified as the reduced-form effect of execution aggressiveness on subsequent forced liquidation by other accounts. The paper does not separately identify the local price path as its exclusive channel.

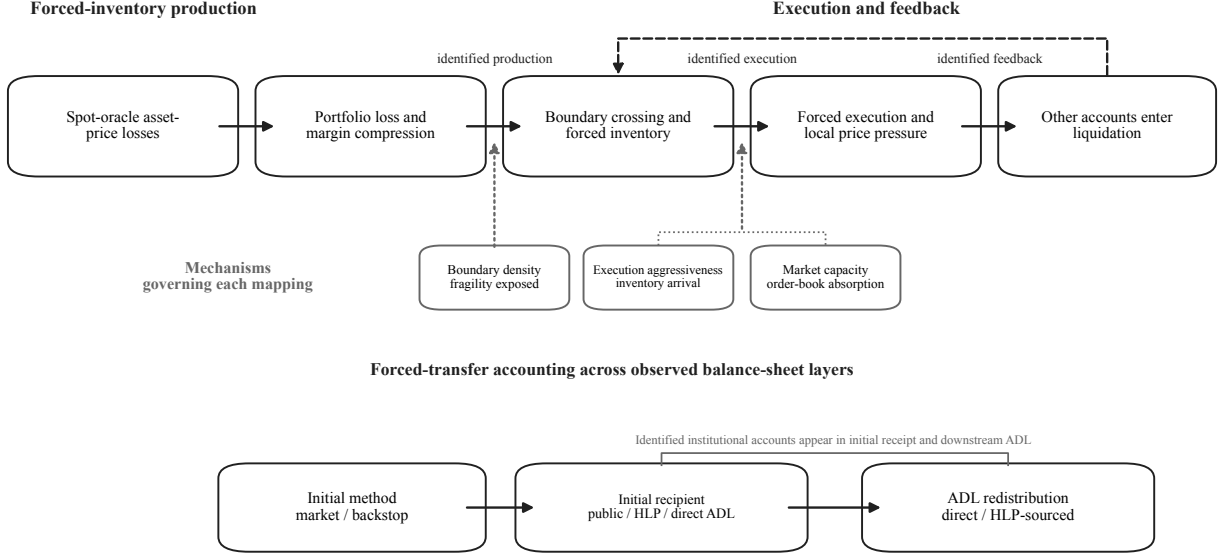


Figure 1: From Margin Constraints to Forced Execution and Risk Transfer

Notes: The figure summarizes the mechanism and the paper’s empirical architecture. The three labeled solid arrows denote links estimated using the October 10 shift-share design and the full-sample \$100,000 discontinuity: forced-inventory production, local execution, and new forced-liquidation onsets by other accounts in the same asset. The dashed return arrow represents potential continuation beyond the estimated downstream response. Dotted connectors link boundary density to the production mapping and execution aggressiveness and market capacity to the execution mapping. The lower panel separately accounts for the initial liquidation method, initial recipient, and downstream ADL. Its bracket indicates that identified institutional accounts appear in initial receipt and downstream ADL. Market denotes order-book execution, backstop denotes transfer to the protocol’s liquidator balance sheet, and ADL denotes automatic deleveraging against selected profitable counterparties. HLP denotes the Hyperliquidity Provider, a community-funded protocol vault.

2.1 Margin Accounts and the Maintenance Boundary

Hyperliquid is a continuous, fully on-chain market for USDC-margined perpetual futures. Traders post collateral and may hold multiple long and short positions within the same account. The platform supports both isolated and cross margin. Isolated positions have dedicated collateral, whereas cross positions share a common collateral pool at the account level. The paper focuses on cross margin because the maintenance constraint is evaluated

against the combined equity and requirements of the cross portfolio.²

At a point in time, account equity consists of posted collateral together with realized and unrealized trading gains and losses and other ledger adjustments, including periodic funding transfers between long and short positions. The maintenance requirement aggregates the requirements generated by the account’s open cross positions. Each position is assigned to an asset-specific maintenance schedule based on its notional size. Some assets face a flat requirement, while tiered schedules impose higher marginal requirements on larger positions. A useful summary of the account’s state is its maintenance surplus: account equity net of the aggregate maintenance requirement. A positive surplus leaves room for prices to move against the portfolio; a smaller surplus places the account closer to forced liquidation. When equity falls below the maintenance requirement, the account becomes liquidatable.

Because cross positions share account equity, losses anywhere in the portfolio reduce the common margin buffer. The relevant shock is therefore the loss on the account’s entire predetermined cross-margin portfolio relative to the equity available to absorb that loss, rather than the return on the position ultimately liquidated. This account-level exposure measure underlies the shift-share design in Section 4.

The maintenance boundary also separates two forms of adjustment. Above the boundary, the trader retains discretion. The account can reduce a losing position, reduce other positions to release margin, deposit collateral, or leave the portfolio unchanged. Dynamic portfolio choice under leverage constraints predicts that the prospect of a future binding constraint can affect trading even before the boundary is reached (Grossman and Vila, 1992; Gârleanu and Pedersen, 2011). These choices motivate the measures of private adjustment introduced in Section 3, including adjustment elsewhere in the portfolio. After the boundary is crossed, the liquidation engine determines the initial quantity and timing of the liquidation order. This transition makes the resulting order flow mechanically linked to the account constraint and substantially less sensitive to the execution price than an ordinary discretionary order.

The same aggregate price decline can consequently produce very different quantities of forced inventory. When few accounts are close to their maintenance boundaries, most losses are absorbed by existing equity or discretionary adjustment. When account equity is densely

²USDC is a dollar-linked stablecoin used to denominate collateral and account balances. Fully on-chain means that trading and account-state transitions are recorded in the platform’s chain state. See the Hyperliquid documentation on margining and liquidations. The main text states the economically relevant rules; margin-tier formulas and the account-state reconstruction are documented in Internet Appendix A.

concentrated near those boundaries, a similar shock moves many accounts into liquidation at approximately the same time. The cross-sectional distribution of margin buffers—rather than average leverage alone—is therefore a state variable governing the potential size of a liquidation cascade.

2.2 Oracle, Mark, and Local Execution Prices

The institutional setting distinguishes three price objects that play different economic roles. The *oracle price* is the platform’s benchmark for the underlying spot asset. Validators form their submissions primarily from spot prices on liquid external exchanges, and the clearinghouse aggregates those submissions into the official oracle. The oracle is therefore distinct from prices formed in Hyperliquid’s perpetual-futures order book.³

The same oracle enters the *mark price*, the platform’s robust estimate of the perpetual contract’s value. The mark combines the oracle with information from Hyperliquid’s book and from external perpetual markets. The mark, rather than the last transaction price, enters unrealized profit and loss, margin calculations, and the liquidation decision. This construction links account valuations to underlying spot conditions, limits the ability of a single local trade to trigger liquidation, and allows persistent local or broader perpetual-market dislocation to affect margin states.⁴

The third object is the local execution price. Market orders generated by the liquidation engine trade against displayed and replenishing liquidity on Hyperliquid. Fill prices, the local midquote, platform-defined impact bid and ask prices that summarize book-implied execution prices at the applicable impact notional, and order-book depth therefore measure the conditions under which forced inventory is executed. These prices can diverge temporarily from both the oracle and the mark, particularly when forced flow is large relative to available depth.

This price hierarchy permits the paper to separate ignition from execution. The shift-share analysis uses minute-by-minute innovations in the spot oracle, interacted with positions

³See the Hyperliquid documentation on the oracle. For assets whose primary spot liquidity resides on Hyperliquid, validator submissions may also incorporate the platform’s spot price. Section 4 examines the timing of oracle innovations relative to local perpetual-market liquidation.

⁴The oracle also serves as the spot benchmark for the premium component of funding. Realized funding payments enter account balances and are incorporated in the account replay, but funding is not the treatment in the shift-share design. See the Hyperliquid documentation on robust price indices and funding.

fixed before the event, to measure losses to predetermined portfolios. It does not use the local perpetual execution price or the separately constructed composite of external perpetual prices as the source shock. The discontinuity analysis instead uses fill and transaction-price outcomes to measure what happens when the resulting forced inventory reaches the local market. Finally, the event anatomy compares the realized mark path with the official spot-oracle benchmark to measure the additional gross position notional held by accounts in margin shortfall along the realized mark path.

2.3 Forced-Liquidation Mechanics

We distinguish the liquidation method, the initial recipient, and each account’s role in subsequent ADL. These related dimensions are not equivalent.

Liquidation begins when mark-to-market account equity falls below the applicable maintenance requirement. The platform first attempts to reduce the distressed positions through orders sent to the ordinary limit-order book. If those orders restore compliance with the maintenance requirement, the trader retains the remaining positions and collateral. Ordinary market liquidation therefore transfers inventory from the constrained account to counterparties willing to trade in the public market.

During the sample period, the initial market order depended discontinuously on pre-liquidation position value. For a liquidatable position larger than \$100,000, the engine initially submitted approximately 20% of the position to the book. Positions below the threshold faced an order for the full position. After a partial liquidation, a 30-second protocol interval governed subsequent liquidation handling; any additional market liquidation during that interval was submitted for the full remaining position. This 30-second interval also defines the episode boundary used to identify new forced-liquidation onsets in Section 5. The threshold therefore changes the initial aggressiveness and timing with which forced inventory reaches the book, even though positions on both sides of the threshold have already entered liquidation. This rule supplies the institutional variation used in Section 5.

Market execution is not always sufficient to restore the account. If book liquidation fails and account equity falls below two-thirds of the maintenance requirement, the protocol transfers the distressed positions and associated margin to the liquidator vault. For a cross-margin account, all cross positions and the remaining cross-margin balance are transferred together;

if the trader has no isolated positions, the transfer leaves the trader with zero account equity. For an isolated position, only that position and its isolated margin are transferred. The backstop is therefore a balance-sheet takeover through the liquidator mechanism rather than another market order sent to the public book. The trader does not recover the maintenance margin, which provides the liquidator vault with a buffer for managing the transferred portfolio. The liquidator vault is a component strategy of the Hyperliquidity Provider (HLP), a community-funded protocol vault, and the resulting liquidation P&L accrues to HLP.⁵

The protocol’s fill metadata label the liquidation method as either **market** or **backstop**, so the method applied to the forced-side account is directly observed. A market-method fill may be received by an ordinary trader or an HLP/system account. A backstop-method fill may transfer the position to the liquidator vault, while a negative account can instead be matched to a trader selected through ADL. The event data therefore classify the forced-side account, the initial recipient, and the ADL role separately rather than treating every counterparty outside the identified HLP/system set as private liquidity.⁶

Automatic deleveraging (ADL) is a separate solvency layer. If an account or isolated position becomes negative, the protocol ranks profitable traders on the opposite side using their profit and leverage and closes selected positions against the underwater account. The same matched trade is then observed from two roles: the underwater account is the forced side, while the profitable counterparty is the ADL-selected side. An ADL-selected counterparty is therefore not voluntary private liquidity. Section 6 uses the source-account identity to distinguish ADL applied directly to accounts outside the identified HLP/system set from ADL applied after an identified HLP/system account itself becomes the forced side.⁷

These stages reveal why closing a distressed account is not the same as removing risk from the system. Market-method fills transfer inventory primarily to public-market counterparties and occasionally to HLP/system accounts. The backstop method ordinarily draws on the liquidator vault’s HLP balance sheet, while a negative account may instead be matched directly to ADL-selected traders. If HLP capacity later becomes impaired, an HLP/system account can itself appear on the forced side of an ADL-linked trade. Resolution can therefore involve a sequence of risk transfers across balance sheets rather than a single terminal trade.

⁵See the official documentation on liquidations and protocol vaults.

⁶The official fill schema records the liquidation method, liquidated user, and mark price. See the Hyperliquid WebSocket data definitions.

⁷See the Hyperliquid documentation on automatic deleveraging.

2.4 A Cross-Margin Liquidation Example

Consider a hypothetical cross-margin account with \$120,000 in equity, \$500,000 in gross position notional, and a \$30,000 maintenance requirement. Its maintenance surplus is therefore \$90,000. The account holds long positions in several assets, including BTC, ETH, and SOL. Because these positions share a common collateral pool, losses on BTC or ETH consume the same buffer that supports the SOL position. Suppose declines in external spot-oracle prices generate losses on the predetermined portfolio and, through the platform’s mark, reduce account equity to \$25,000. Equity is now below the \$30,000 maintenance requirement, so the account becomes liquidatable. In this sequence, oracle-price movements measure the external asset-price shock, while mark-valued equity determines whether the account crosses the liquidation boundary.

Suppose the SOL position has a pre-liquidation value near the \$100,000 policy threshold. Compare two otherwise similar liquidated positions valued at \$99,000 and \$101,000. Both have already entered market liquidation. Below the threshold, the engine initially targets the full position; above it, the engine initially targets approximately 20% of the position. The rule therefore changes the amount of forced-sale inventory submitted at entry, not whether liquidation occurs. When displayed and replenishing bid liquidity is limited, the full-position order exposes more inventory to the book immediately and can produce worse execution and greater short-run transaction-price pressure.

At \$25,000 in equity, the account is liquidatable but remains above the backstop condition of two-thirds of its \$30,000 maintenance requirement, or \$20,000. If book execution reduces the maintenance requirement sufficiently to restore a positive surplus, the trader retains the remaining positions and collateral. If book liquidation fails and the account subsequently falls below \$20,000, the protocol can transfer the cross-margin portfolio and its remaining balance to the liquidator vault. The transfer places the positions on an HLP balance sheet rather than eliminating the underlying risk. If that HLP/system account later becomes impaired, it can itself appear as the forced side of an ADL-linked trade, with profitable traders selected on the opposite side for involuntary deleveraging.

The example also distinguishes the three classifications used throughout the paper. The *liquidation method* records how the distressed account is processed; the *initial recipient* identifies the first balance sheet receiving the position; and the *ADL role* identifies the account subsequently forced and the counterparty selected. A market-method liquidation can be

received by either an ordinary trader or an HLP account, and an account in the identified institutional set can later enter the forced sequence.

2.5 Conceptual Mechanism

The framework separates three identified mappings—the production of forced inventory, its execution in the market, and its feedback into subsequent same-asset liquidation—from downstream transfer across balance sheets, which is measured through event-path and flow accounting. It is built around observable balance sheets, orders, prices, and counterparties. Its purpose is to discipline the empirical sequence and clarify which links are identified by quasi-experimental variation and which are measured through event-path and flow accounting.

The production mapping begins with heterogeneous exposure to an asset-price shock. Accounts enter the event with different signed positions, equity, and maintenance surpluses. An oracle-price innovation produces a larger loss for an account whose frozen portfolio is more exposed to the affected assets. Whether that loss becomes forced supply depends on the account’s initial margin buffer. At the system level, the interaction between shock size and boundary density determines how many accounts cross into liquidation and how much initial inventory the engine must execute.

The execution mapping begins after boundary crossing. Forced inventory is comparatively price insensitive over the short horizon in which the platform must restore margin compliance. Its immediate market effect nevertheless depends on how it is executed. A large initial order consumes more displayed and latent liquidity than a sequence of smaller orders; the same order produces greater dislocation when depth is thin or when potential buyers have already committed balance-sheet capacity elsewhere. This role for financially constrained and slow-moving risk-bearing capital is central to theories of liquidity provision under stress (Gromb and Vayanos, 2002; Duffie, 2010). Execution aggressiveness and contemporaneous market capacity thus map a given balance-sheet loss into execution costs and local price pressure.

Local dislocation can then feed back into account states. The mark incorporates both external and local perpetual-market information, and arbitrage can transmit venue-specific pressure to prices elsewhere. A sufficiently persistent adverse price path reduces unrealized

profit and loss and compresses the maintenance surpluses of accounts that survived the initial shock. If additional accounts cross the boundary, they generate a later wave of forced inventory. Section 5 identifies the local reduced-form effect of execution aggressiveness on subsequent same-asset liquidation, while Section 6 measures its event-level counterpart along the realized valuation path.

Downstream risk transfer depends on the availability of new risk-bearing capacity. Ordinary counterparties may absorb inventory directly, while the liquidator vault provides a backstop when the book cannot restore account solvency. If losses still produce negative equity, ADL redistributes positions and losses to profitable traders. These mechanisms can stop the mechanical feedback, but they do so by moving risk to other balance sheets rather than by making the inventory disappear.

The framework yields three empirical predictions. *Prediction 1 (forced-inventory production)* is that, conditional on initial margin state and exposure to the common market decline, a larger loss generated by predetermined portfolio exposure causes earlier entry into liquidation and more forced inventory at entry. Section 4 tests this prediction by freezing account portfolios before the October 10 cascade and interacting them with subsequent relative spot-oracle innovations.

Prediction 2 (forced execution) is that, among positions already entering liquidation, less aggressive initial execution reduces adverse execution costs and immediate directional price pressure. Section 5 tests this prediction using the discontinuity in the initial liquidation-order share at \$100,000. The estimand is the local effect of liquidation policy among positions already entering market-method liquidation near the threshold, where initial execution aggressiveness changes sharply.

Prediction 3 (feedback) is that, among positions already entering liquidation, more aggressive initial execution increases the amount and breadth of subsequent forced liquidation by other accounts in the same asset. Section 5 tests this prediction using the same discontinuity, with new forced-liquidation onsets by other accounts as the outcome.

Two system implications complete the mechanism. First, the local feedback identified in Section 5 can aggregate through boundary density and market capacity; Section 6 measures its event-level counterpart. Second, the resulting positions and losses must be transferred to private or institutional balance sheets or redistributed through ADL. Section 3 documents boundary density and private adjustment across snapshot windows, while Section 6 docu-

ments their event-level counterparts in the October 10 event. The empirical analysis therefore moves from the distribution of margin fragility, to the identified production, execution, and feedback links, and finally to system-level exposure and downstream risk transfer.

3 Data and Broader Evidence on Margin Fragility

We first establish the balance-sheet conditions that govern how much forced inventory a price shock can produce. The broader panel serves two purposes. It shows how margin pressure is distributed across accounts and documents how adjustment changes as losses interact with initial leverage. The central pattern is a transition from private position reduction to severe forced liquidation in the upper tail of the margin-pressure distribution, consistent with account-level evidence that trading changes as investors approach margin constraints (Kahraman and Tookes, 2017; Bian et al., 2026). The high-frequency October 10 replay then measures the same state variable directly through each eligible cross-margin account’s maintenance surplus. Together, the two samples connect the ordinary adjustment of leveraged portfolios to the concentrated fragility that precedes a liquidation cascade.

3.1 Linked Data Architecture

The empirical analysis links account balance sheets, positions, prices, orders, executions, and counterparties on a common clock. Account snapshots report collateral, the platform’s account-value field, which we use as account equity, and open positions by asset and margin mode. We combine these snapshots with the platform’s oracle and mark histories to value positions and reconstruct maintenance requirements. The resulting account states contain initial equity, gross and net exposure, effective leverage, maintenance surplus, and the composition of each cross-margin portfolio.

We next link these states to the complete sequence of position changes. Orders and fills identify the asset, side, quantity, execution price, account, and whether the trade is ordinary or protocol-directed through the observed market, backstop, and ADL routes. The raw records label protocol-directed liquidation sourced from a trader account as “user liquidation”; despite the label, this is forced rather than user-initiated execution. This classification separates voluntary position adjustment from protocol-directed execution. Matching

the fills back to beginning-of-window positions allows us to measure how much of an initial position is privately reduced, whether the reduction persists through the end of the window, when an account first enters liquidation, and how much forced inventory the account produces. For the October 10 analysis, limit-order-book snapshots add spreads, impact prices, and depth, while counterparty identifiers trace which accounts receive market-routed and backstop-routed inventory.

The value of the data comes from these links. A position can be placed inside its account-wide collateral pool, exposed to the contemporaneous oracle-price path, classified as privately or forcibly reduced, matched to its execution conditions, and followed to the balance sheet that acquires it. The observation count reflects this depth rather than repeated price records alone. The broad panel applies this architecture across the full panel of snapshot windows, while the October 10 replay expands it to account–asset–minute and fill-level resolution.

3.2 Empirical Samples and Measurement

Table 1 summarizes the paper’s main estimation samples. The broad panel covers 215 approximately daily snapshot-to-snapshot windows from September 28, 2025 through April 30, 2026. Each window begins with the observed portfolio and account state and ends at the next daily snapshot. Oracle returns are aligned to the actual endpoints of each window. This construction measures the price change applied to the position that was present at the beginning of the window and preserves the account’s initial balance sheet before any within-window adjustment.

The broad-panel analysis is restricted to ordinary user-controlled accounts; protocol or system accounts, HLP-related accounts, and vault accounts are excluded.⁸ Both broad-panel samples require at least \$100 of initial account equity, positive cross-margin notional, and finite positive cross-margin leverage. The private-adjustment sample further requires a positive initial target position, matched endpoint benchmark prices, an observed within-window leverage rank, and complete source, direction, and trading-path reconstruction. Its observation unit is the account–asset–window. The estimation sample contains 29,488,255 observations spanning 190,855 accounts, 195 assets, and all 215 windows. Collapsing these

⁸Vaults are pooled strategy accounts rather than ordinary user-controlled trading accounts. Internet Appendix A documents the account classification and exclusions.

positions to the account–window level yields 5,920,216 unique account–windows for the initial balance-sheet statistics reported in Panel A of Table 1.

Throughout the broad-panel analysis, i indexes accounts, j indexes assets, and w indexes snapshot-to-snapshot windows; endpoint subscripts 0 and 1 denote the beginning and end of a window. Let $q_{ijw,0}$ and $q_{ijw,1}$ be the signed position quantities at those endpoints, with positive values denoting long positions and negative values denoting short positions, and define $Q_{ijw,0} = |q_{ijw,0}|$ and $Q_{ijw,1} = |q_{ijw,1}|$. For each initial position, we classify whether its signed exposure loses value under the window’s oracle return. Let D_{ijw}^{priv} denote the cumulative amount of the original-direction exposure reduced through non-forced trading before the first forced execution, or through the window end if no forced execution occurs, capped at $Q_{ijw,0}$. The main private-adjustment outcome is

$$\text{PersistentPrivateReduction}_{ijw} = \frac{\min \left\{ D_{ijw}^{\text{priv}}, (Q_{ijw,0} - Q_{ijw,1})_+ \right\}}{Q_{ijw,0}}. \quad (1)$$

The minimum retains only the portion of private reduction supported by the endpoint position change. A temporary reduction that is fully reversed through re-entry therefore receives a value of zero, while a reduction that remains in place contributes in proportion to the initial position.

The forced-liquidation sample aggregates the same ordinary-account positions to the account–window level and applies the same initial balance-sheet restrictions. It additionally requires benchmark prices for at least 80% of initial cross-margin notional and complete loss and liquidation measures. The resulting sample contains 5,887,276 account–window observations from 190,490 cross-margin accounts across the same 215 windows. The account’s oracle-implied portfolio loss is

$$\text{Loss}_{iw}^O = \frac{\max \left\{ -\sum_j q_{ijw,0} P_{jw,0}^O R_{jw}^O, 0 \right\}}{E_{iw,0}}, \quad (2)$$

where $P_{jw,0}^O$ is asset j ’s beginning-of-window oracle price in dollars per unit, $P_{jw,1}^O$ is its endpoint oracle price, and $R_{jw}^O = P_{jw,1}^O / P_{jw,0}^O - 1$ is its oracle return over window w . Thus, $q_{ijw,0} P_{jw,0}^O$ is account i ’s signed beginning-of-window dollar exposure to asset j , and $q_{ijw,0} P_{jw,0}^O R_{jw}^O$ is the corresponding oracle-implied position profit or loss holding the initial

quantity fixed. Finally, $E_{iw,0}$ is account i 's equity at the beginning of window w . The principal forced-liquidation outcome is forced-liquidation notional divided by initial equity. Scaling losses and liquidation by equity maps the price path into the balance-sheet resources available to absorb it.

The broad panel measures initial margin pressure using cross-margin leverage. We rank eligible accounts by initial cross-margin leverage within each window and define $HighLev_{iw}$ as the highest two deciles. This within-window rank provides a common measure of elevated margin pressure across dates and market conditions. The October 10 replay supplies the minute-level account states and reconstructed distance to the maintenance boundary defined in Section 3.3.

Panel B of Table 1 introduces the linked-portfolio sample used to measure adjustment across positions sharing the same collateral pool. It contains 27,843,280 account–target–asset–window observations from 112,748 accounts, 195 target assets, and 215 windows. Panel C reports the October 10 first-liquidation risk set: accounts enter at 20:45 UTC and remain until their first eligible forced liquidation or 21:12 UTC. It contains 1,809,043 account–minute observations for 69,137 accounts across 28 outcome minutes, with portfolio exposures and risk states frozen at 20:44 UTC. Panel D reports 3,562 long-position market-liquidation events within \$20,000 of the \$100,000 policy threshold, drawn from 2,813 accounts, 141 assets, and 200 UTC dates. Presenting the samples together makes the empirical sequence visible: the broad panel establishes margin pressure and adjustment, the linked sample measures portfolio-wide reallocation, the risk set identifies entry into liquidation, and the policy sample identifies the local effect of execution aggressiveness.

Table 1: Data, Samples, and Summary Statistics

Variable	Unit	Mean	Std. dev.	Median	P90	Observations
<i>Panel A. Broad daily-window account and position characteristics</i>						
Private-adjustment sample: account-window						
Initial account equity	\$000	91.167	1,229.440	1.800	62.735	5,920,216
Initial gross cross-margin notional	\$000	236.262	4,283.703	3.377	128.006	5,920,216
Cross-margin leverage	ratio	3.687	15.905	1.927	8.787	5,920,216
Oracle-implied portfolio loss / initial equity	%	5.311	16.160	0.045	14.393	5,920,216
Private-adjustment sample: account-asset-window						
Initial position notional	\$000	47.660	1,362.806	0.163	16.774	29,488,255
Persistent private reduction	%	11.114	28.236	0.000	54.839	29,488,255
Forced-liquidation sample: account-window						
Any forced liquidation during window	%	1.854	13.490	0.000	0.000	5,887,276
User-liquidation notional / initial equity	%	28.657	379.173	0.000	0.000	5,887,276
<i>Panel B. Linked-portfolio adjustment</i>						
Linked-portfolio sample: account-target-asset-window						
Number of non-target source assets	count	41.565	43.824	45.000	100.000	27,843,280
Leave-target-out portfolio loss / initial equity	%	1.834	7.238	0.000	4.451	27,843,280
Target-position reduction persisting to window end	%	10.429	27.285	0.000	49.306	27,843,280
<i>Panel C. October 10 first-liquidation risk set</i>						
First-liquidation risk set						
Account equity at 20:44 UTC	\$000	53.102	1,327.523	0.528	23.419	1,809,043
Gross cross-margin notional at 20:44 UTC	\$000	183.829	5,358.823	1.043	60.897	1,809,043
Cross-margin leverage at 20:44 UTC	ratio	4.021	5.356	2.233	9.297	1,809,043
Distance to maintenance boundary at 20:44 UTC	ratio	0.855	0.153	0.913	0.980	1,808,163
Current external portfolio loss / frozen equity	%	0.056	1.625	0.000	0.985	1,809,043
First liquidation begins in current minute	%	0.489	6.975	0.000	0.000	1,809,043
Entry-minute forced inventory / frozen gross notional	%	0.385	5.863	0.000	0.000	1,809,043
<i>Panel D. \$100,000 partial-liquidation policy sample</i>						
Long market-liquidation events, $\pm \$20k$						
Pre-liquidation position value	\$000	99.739	11.276	99.697	115.452	3,562
Initial liquidation-order share	%	51.931	38.654	23.106	100.000	3,562
Adverse execution slippage	bps	39.368	140.134	5.004	74.532	3,562
Adverse transaction-price change at 10 seconds	bps	21.134	115.190	2.116	65.130	3,490
Maximum adverse transaction-price move, 0–10s	bps	58.058	165.984	11.095	120.536	3,524
Downstream-outcome sample						
Isolation-restricted 300-second subsample	events					2,067
New liquidation notional, 30 seconds	\$m	0.268	0.857	0.021	0.569	2,067
New liquidation notional, 300 seconds	\$m	1.069	4.208	0.146	2.546	2,067
Newly liquidated other accounts, 30 seconds	count	23.869	54.929	6.000	57.400	2,067
Newly liquidated other accounts, 300 seconds	count	80.462	175.643	24.000	206.000	2,067

Notes: Statistics are computed from the frozen estimation samples described in Section 3.2. The Observations column reports the number of nonmissing finite values for each displayed variable, not the total size of the corresponding estimation sample. Row-level counts can therefore differ when a variable requires additional price or state support.

Panel A: Beginning-of-window balance-sheet variables are collapsed to one observation per account–window so that accounts with broader portfolios do not receive greater weight; position notional and persistent private reduction remain at the account–asset–window level. Initial equity and gross cross-margin notional are measured at the opening snapshot, and leverage is their ratio. Oracle-implied portfolio loss holds signed initial positions fixed, applies endpoint-to-endpoint oracle returns, retains the loss component, and scales it by initial equity. Private reduction excludes forced execution and counts only reductions in the initial direction that remain visible in the endpoint position. Forced-liquidation notional cumulates protocol-directed reductions recorded under the raw user-liquidation label during the window; its ratio to initial equity is not capped.

Panel B: The unit is an account–target–asset–window. The number of source assets counts non-target positions in the same beginning-of-window cross-margin portfolio. Leave-target-out portfolio loss applies oracle returns to those source positions, retains their combined loss, and scales it by initial account equity. Target-position reduction uses the same persistent, non-forced definition as in Panel A.

Panel C: Portfolio exposures and risk states are fixed at 20:44 UTC and followed in the first-liquidation risk set from 20:45 through 21:12 UTC. Distance is reconstructed maintenance surplus scaled as in equation (3); smaller values indicate thinner buffers. Current external portfolio loss combines frozen signed exposures with the asset-specific oracle innovation observed at the start of the outcome minute after removing the common oracle-market component. The first-liquidation indicator equals one in the minute in which an account first enters forced liquidation. The entry-minute forced-inventory rate divides forced-liquidation notional in that minute by frozen gross cross-margin notional and is capped at one.

Panel D: The unit is a long position entering market-method liquidation within \$20,000 of the \$100,000 policy cutoff. Position value equals absolute pre-liquidation quantity times the pre-event mark. The initial liquidation-order share divides cumulative initial-order quantity by the pre-liquidation position and is capped at one. Execution slippage is measured adversely relative to the pre-event mark. Transaction-price outcomes are measured relative to the last valid pre-event trade; their smaller observation counts reflect unavailable post-event trade-price support. The downstream rows use the common 2,067-event sample with complete 300-second support and no earlier eligible focal event in the same asset during the preceding 300 seconds. They exclude the focal account, count each other account once at its first new market-liquidation onset following more than 30 seconds without a market- or backstop-liquidation fill, and sum the initial-order notionals generated at those onsets. Position values are reported in thousands, new-liquidation notional in millions, ratios in their natural units, and fractions and indicator means in percentage points.

3.3 The Distribution of Margin Fragility

Figure 2 presents two views of margin fragility. Panel A pools eligible account–windows from all 215 windows and assigns equal-sized deciles according to realized margin pressure, defined as the interaction between oracle-implied portfolio loss and initial cross-margin leverage. The deciles are formed across the pooled sample rather than separately within each window. Unlike the within-window leverage rank used to define $HighLev_{iw}$, this descriptive sorting retains both cross-account and cross-window variation in realized pressure. Forced-liquidation probabilities are account–window means. To calculate private-reduction probabilities, we map each account–window’s decile to its eligible initial positions and average an indicator for positive persistent reduction at the account–asset–window level; the private-reduction series is therefore position-weighted. Private reduction is already common in the lower deciles, ranging between 15% and 19%, and rises sharply to 29.6% in the highest-pressure decile. Forced liquidation displays a more concentrated response. Its probability is 0.64% in the lowest decile, remains below 1% through most of the distribution, and reaches 13.2% in the highest decile.

The contrast between the two series reveals the adjustment sequence. Private trading absorbs moderate pressure across a wide range of account states. Forced liquidation becomes quantitatively important when a large oracle-implied loss meets an already leveraged balance sheet. The upper tail therefore governs how much of a common market decline is converted into price-insensitive inventory.

To measure this boundary directly during October 10, we reconstruct account balance sheets at one-minute frequency. We initialize the replay from the full-account snapshot at 20:04:54 UTC and advance positions and balances through subsequent position-changing fills, funding payments, and other ledger events in timestamp order. Contemporaneous oracle and mark-price histories value each state and determine the applicable maintenance requirement. The resulting master replay contains 8,967,821 account–minute states and 21,397,216 account–asset–minute states. Figure 2B uses eligible cross-margin states with complete maintenance inputs. Table 1, Panel C instead reports the narrower first-liquidation risk set used in Section 4: portfolios and risk states are frozen at 20:44 UTC, and each account is followed from 20:45 until its first eligible liquidation or 21:12 UTC. Appendix A describes the replay inputs, sample coverage, and validation.

For each eligible cross-margin account i in minute t , we define the reconstructed distance

to the maintenance boundary as

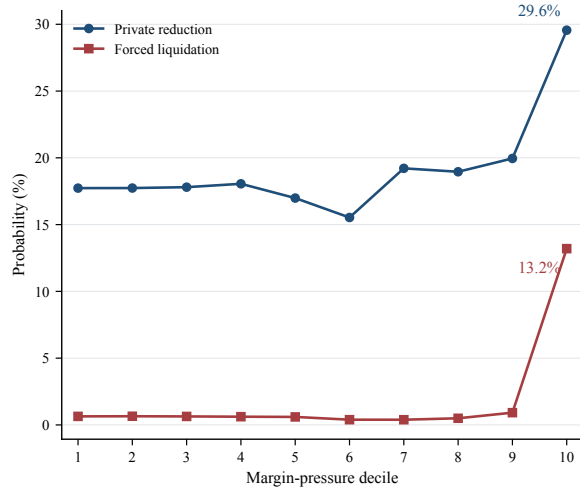
$$Distance_{it}^{Cross} = \frac{E_{it} - \sum_j MM_{ijt}}{\max\{|E_{it}|, 0.01G_{it}, 1\}}, \quad (3)$$

where E_{it} is replayed account equity, MM_{ijt} is position j 's maintenance-margin contribution under the applicable maintenance-margin schedule described in Section 2.1 and Appendix A, and G_{it} is gross cross-margin notional. The numerator is the account's dollar maintenance surplus. A smaller value denotes a thinner margin buffer, and a negative value denotes a reconstructed maintenance shortfall. The positive denominator places accounts of different sizes on a comparable scale without changing whether the maintenance boundary has been crossed.

Panel B pools the eligible October 10 cross-margin account–asset–minute states and assigns deciles according to $Distance_{it}^{Cross}$ across the event sample rather than separately within each minute. The account-level distance is mapped to each associated position state, with decile 10 containing the smallest buffers, and the outcome records whether that position experiences forced liquidation over the following one, five, or fifteen minutes. The curves are strongly nonlinear at every horizon. At five minutes, liquidation probability rises from 0.124% in the least fragile decile to 8.211% in the most fragile decile, a 66.3-fold increase. The corresponding one- and fifteen-minute probabilities rise from 0.030% to 2.116% and from 0.341% to 21.052%. Across all three horizons, positions held by accounts closest to the maintenance boundary enter liquidation sooner and at much higher rates throughout the subsequent event window.

These distributions highlight the balance-sheet state variable that organizes the remainder of the paper. Market-wide price changes determine the aggregate loss, but the cross-sectional density of maintenance surpluses determines how many accounts cross into forced execution. A system with substantial equity buffers can absorb the same price move through private adjustment. A system with many accounts concentrated near the boundary produces a discrete increase in forced inventory.

A. Broad daily-window gradient



B. Exact cross-margin distance on October 10

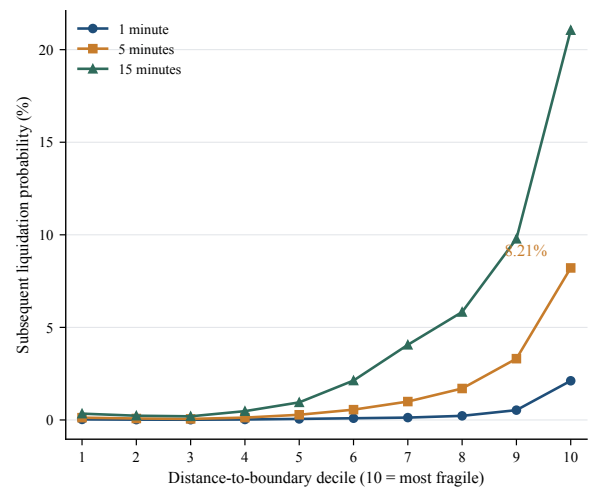


Figure 2: Margin Fragility and the Transition to Forced Liquidation

Notes: Panel A pools eligible account–windows across 215 approximately daily windows and forms equal-sized deciles of oracle-implied portfolio loss interacted with initial cross-margin leverage; the deciles are not recomputed within each window. Forced liquidation is averaged at the account–window level. Private reduction is averaged across eligible initial positions after mapping the account–window decile to the account–asset–window panel. Panel B pools eligible October 10 cross-margin account–asset–minute states with complete maintenance inputs and forms deciles of the reconstructed account-level distance in equation (3); the deciles are not recomputed within each minute, and decile 10 contains the smallest margin buffers. The outcome is whether the associated position experiences forced liquidation over the indicated forward horizon.

3.4 Private and Portfolio-Wide Adjustment

We next estimate how initial margin pressure changes private adjustment. For account i , asset j , and window w , the main specification is

$$\begin{aligned} \text{PersistentPrivateReduction}_{ijw} = & \beta_1 \text{HighLev}_{iw} + \beta_2 \text{LosingLeg}_{ijw} \\ & + \beta_3 \text{HighLev}_{iw} \times \text{LosingLeg}_{ijw} \\ & + \beta_4 \log \text{Notional}_{ijw,0} + \alpha_{jw} + \mu_i + \varepsilon_{ijw}, \end{aligned} \quad (4)$$

where $\text{PersistentPrivateReduction}_{ijw}$ is the persistent private-reduction share defined in equation (1), HighLev_{iw} identifies accounts in the top two deciles of initial cross-margin leverage within window w , LosingLeg_{ijw} indicates that the beginning-of-window signed position loses under the oracle return, and α_{jw} denotes asset-by-window fixed effects. Column (2) additionally includes account fixed effects.

Panel A of Table 2 reports the interaction coefficient. Losing positions held by *HighLev* accounts have a 0.608-percentage-point higher persistent private-reduction rate. Relative to the 11.11% dependent-variable mean, the estimate represents a 5.5% increase. With account fixed effects, the estimate is 0.219 percentage points. The estimate is similar when October 10 or the October 9–11 interval is excluded (Internet Appendix Table B2).

The account-level specification examines the severity of forced adjustment:

$$\begin{aligned} \text{LiquidationSeverity}_{iw} = & \theta_1 \text{Loss}_{iw}^O + \theta_2 \text{HighLev}_{iw} \times \text{Loss}_{iw}^O + \theta_3 \text{Leverage}_{iw,0} \\ & + \theta_4 \text{Leverage}_{iw,0} \times \text{Loss}_{iw}^O + \theta_5 \log E_{iw,0} + \delta_w + \mu_i + u_{iw}, \end{aligned} \quad (5)$$

where $\text{LiquidationSeverity}_{iw}$ is cross-margin forced-liquidation notional scaled by initial equity and δ_w denotes window fixed effects. Column (2) additionally includes account fixed effects. The coefficient of interest, θ_2 , measures the incremental sensitivity of liquidation severity to oracle-implied portfolio losses among *HighLev* accounts, beyond the smooth loss gradient associated with continuous initial leverage.

Panel B of Table 2 reports θ_2 . A ten-percentage-point increase in oracle-implied portfolio loss is associated with additional liquidation notional equal to 50.614 percentage points of initial equity for *HighLev* accounts, beyond the smooth loss gradient associated with continuous initial leverage. With account fixed effects, the corresponding estimate is 47.464

percentage points. The magnitude remains similar after excluding October 10 or the October 9–11 interval. Margin pressure therefore predicts economically large quantities of inventory transferred through forced execution.

Table 2: Margin Pressure, Private Adjustment, and Forced Liquidation

	(1)	(2)
<i>Panel A. Persistent private reduction</i>		
<i>HighLev</i> × losing leg	0.608*** (0.168)	0.219* (0.127)
Asset–window fixed effects	Yes	Yes
Account fixed effects	No	Yes
Observations	29,488,255	29,488,255
R^2	0.013	0.008
<i>Panel B. Forced-liquidation notional scaled by initial equity</i>		
<i>HighLev</i> × oracle-implied portfolio loss	50.614*** (5.229)	47.464*** (4.560)
Window fixed effects	Yes	Yes
Account fixed effects	No	Yes
Observations	5,887,276	5,887,276
R^2	0.150	0.107

Notes: Panel A is estimated at the account–asset–window level for cross-margin accounts with initial equity of at least \$100, positive cross notional, and finite positive cross leverage. The dependent variable is persistent private reduction, and coefficients are reported in percentage points. Column (1) absorbs asset–window fixed effects; column (2) additionally absorbs account fixed effects. Both specifications control for the *HighLev* indicator (the top two within-window leverage deciles), the losing-leg indicator, and log initial position notional.

Panel B is estimated at the account–window level. The dependent variable is forced-liquidation notional divided by initial equity and is reported in percentage points. The focal coefficient is the incremental oracle-implied-loss slope for *HighLev* accounts after controlling for continuous initial leverage and its interaction with oracle-implied loss; estimates are scaled to a ten-percentage-point portfolio loss. Both columns include oracle-implied portfolio loss, continuous initial leverage, their interaction, and log initial account equity. Column (2) additionally absorbs account fixed effects.

The reported R^2 is the within R^2 after absorbing the listed fixed effects. Standard errors in parentheses are two-way clustered by account and asset–window in Panel A and by account and window in Panel B. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Portfolio-wide adjustment. To test whether margin pressure changes positions beyond the asset in which the loss originates, we estimate

$$\begin{aligned} \text{PersistentPrivateReduction}_{ijw} = & \lambda \text{LinkedLoss}_{iw}^{-j} + \gamma \text{OwnLoss}_{ijw} + \rho \text{HighLev}_{iw} \\ & + \kappa \log \text{Notional}_{ijw,0} + X'_{ijw} \Gamma + \alpha_{jw} + \varepsilon_{ijw}. \end{aligned} \quad (6)$$

$\text{LinkedLoss}_{iw}^{-j}$ is the oracle-implied loss on all initial cross-margin positions other than target asset j , scaled by initial equity; OwnLoss_{ijw} is the target position’s own oracle-implied loss. The vector X_{ijw} contains source-portfolio composition, continuous initial leverage, and benchmark source-coverage controls, while α_{jw} denotes target-asset-by-window fixed effects. The coefficient of interest, λ , measures whether losses elsewhere in the shared collateral pool predict persistent reduction of the target position, conditional on the target’s own loss and target-asset-by-window conditions.

In the controlled specification, the estimate of λ is 13.807 when the dependent variable is expressed in percentage points. Equivalently, a ten-percentage-point increase in leave-target-out portfolio loss is associated with a 1.381-percentage-point increase in persistent private reduction. In the subsample in which the target position itself does not lose value, the baseline estimate is 19.502, corresponding to 1.950 percentage points for a ten-percentage-point linked loss (Internet Appendix Table B3, Panel A). Losses originating elsewhere in the account therefore predict adjustment of the target position, consistent with portfolio-wide risk reduction within accounts. This account-level channel is related to evidence that common ownership and correlated trading needs transmit balance-sheet shocks across assets (Greenwood and Thesmar, 2011; Bian et al., 2026).

The broader evidence documents the balance-sheet conditions associated with the production of forced inventory. Fragility is concentrated in the tail of the margin-state distribution, private adjustment begins within and across positions, and severe losses convert high initial leverage into large quantities of forced inventory. Section 4 next freezes these pre-existing states before the October 10 cascade and uses cross-asset oracle innovations to identify when portfolio losses move latent fragility across the maintenance boundary and generate initial forced inventory.

4 External Portfolio Shocks and the Production of Forced Inventory

Section 3 shows that forced liquidation is concentrated among accounts with high initial leverage and small margin buffers. We now estimate how post-freeze asset-specific oracle innovations convert this latent fragility into realized liquidation entry and initial forced inventory. The design compares accounts observed in the same minute and in similar predetermined margin states, but whose frozen portfolios load differently on those innovations.

4.1 Predetermined Portfolios and External Loss Exposure

We freeze each eligible cross-margin account’s portfolio and balance-sheet state at 20:44 UTC on October 10. The frozen state contains positions by asset, account equity, gross and net exposure, leverage, portfolio breadth, and reconstructed distance to the maintenance boundary. We then observe asset-level oracle innovations and account liquidations from 20:45 through 21:12, immediately before the core cascade begins at 21:13. An account remains in the risk set until its first eligible forced liquidation. Liquidation outcomes are measured from fills occurring after the oracle-price snapshot at the beginning of each minute. Because the outcome is the production of forced inventory before its execution route is determined, both market-method and backstop-method forced liquidations count as entry into forced execution. Section 5 isolates market-method liquidations when estimating local execution effects; ADL is recorded separately.

This construction produces 1,809,043 at-risk account-minutes from 69,137 accounts. The window contains 8,845 first-liquidation events and \$473.81 million of forced inventory executed during the first minute of those events. Freezing the portfolio before the analysis window fixes the assets through which each account is exposed, while the first-liquidation risk set aligns the price innovation that tightens the account with the first inventory transferred through forced execution.

Let V_{ia0} denote account i ’s signed dollar exposure to source asset a at 20:44 and let E_{i0} denote frozen account equity. The predetermined signed exposure weight is

$$s_{ia0} = \frac{V_{ia0}}{E_{i0}}. \quad (7)$$

The weight is leverage-scaled rather than a conventional portfolio share: it may be negative or exceed one and need not sum to one across assets. A positive value represents a long exposure and a negative value represents a short exposure. Because positions are scaled by frozen equity, the resulting portfolio innovation measures the balance-sheet loss generated by a given asset-price movement.

For each source asset, we separate its one-minute oracle return from the common market movement. Using the 60 days before the portfolio freeze, we estimate α_a^{5m} and β_a from five-minute oracle returns and a lagged-open-interest-weighted leave-one-out oracle-market factor. We hold these coefficients fixed during the event window and convert the five-minute intercept to one-minute units by dividing it by five. The relative oracle innovation is

$$\varepsilon_{at} = r_{at}^O - \left(\frac{\alpha_a^{5m}}{5} + \beta_a f_{-a,t} \right), \quad (8)$$

where r_{at}^O is the one-minute oracle return and $f_{-a,t}$ is the contemporaneous one-minute oracle-market factor constructed without asset a . Interacting these source innovations with the frozen portfolio yields account i 's current externally induced loss:

$$X_{it} = - \sum_a s_{ia0} \varepsilon_{at}. \quad (9)$$

A positive X_{it} is a loss as a fraction of frozen equity. Two accounts observed in the same minute can therefore receive different portfolio losses because their pre-event positions load differently on the same collection of asset-specific oracle innovations.

This construction follows the quasi-experimental shock-exposure logic of shift-share designs: conditional on predetermined exposure weights and the included controls, identification is tied to the source innovations that differentially affect otherwise comparable accounts (Borusyak et al., 2022).

We examine two outcomes. The first records entry into forced liquidation,

$$FirstLiquidation_{it} = \mathbf{1}\{T_i \in [t, t + 1)\}, \quad (10)$$

where T_i is account i 's first eligible forced-liquidation time. The second measures total initial

forced inventory across the at-risk population:

$$\begin{aligned} EntryMinuteForcedInventoryRate_{it} = FirstLiquidation_{it} \\ \times \min \left\{ \frac{EntryMinuteLiquidationUSD_i}{InitialCrossGrossNotional_{i0}}, 1 \right\}. \end{aligned} \quad (11)$$

The numerator sums the account’s forced-liquidation notional during its entry minute. Defined for every at-risk account-minute, the outcome equals zero in the absence of liquidation entry and measures total initial forced inventory per unit of frozen cross-margin gross notional. The bound prevents a sequence of within-minute fills from producing a rate above one.

4.2 Empirical Specification

For outcome $k \in \{FirstLiquidation, EntryMinuteForcedInventoryRate\}$, we estimate

$$\begin{aligned} Y_{it}^k = \lambda_k X_{it} + \gamma_k L_{i,t-1}^{rel} + \theta_{1k} C_{it}^{new} + \theta_{2k} C_{i,t-1}^{lag} + B_i' \Gamma_k + \delta_t \\ + \phi_{q_E(i),t}^E + \phi_{q_L(i),t}^L + \phi_{q_D(i),t}^D + u_{it}^k. \end{aligned} \quad (12)$$

The variable $L_{i,t-1}^{rel}$ is the relative portfolio loss accumulated through the preceding minute. The terms C_{it}^{new} and $C_{i,t-1}^{lag}$ capture current and accumulated portfolio losses generated by the common oracle-market factor. The vector B_i contains frozen account size, net-to-gross exposure, and portfolio breadth, and δ_t denotes minute fixed effects.

Here, $q_E(i)$ and $q_L(i)$ denote account i ’s frozen-equity and frozen-leverage quintiles, while $q_D(i)$ denotes the decile of its reconstructed distance to the maintenance boundary; all three groups are assigned once at 20:44. The terms $\phi_{q_E(i),t}^E$, $\phi_{q_L(i),t}^L$, and $\phi_{q_D(i),t}^D$ are, respectively, equity-group-by-minute, leverage-group-by-minute, and distance-group-by-minute fixed effects. Equation (12) consequently compares accounts observed in the same minute and in similar predetermined risk states, but whose frozen portfolios load differently on contemporaneous asset-specific oracle innovations. In both panels of Table 3, column (1) omits account fixed effects and column (2) additionally absorbs account fixed effects μ_i . Because B_i is fixed at 20:44, its components are absorbed rather than separately estimated in the account-fixed-effect specifications. The coefficient λ_k measures how a larger current external

loss changes first-liquidation entry and the entry-minute forced-inventory rate, conditional on the account’s accumulated loss and the common market decline.

Standard errors are clustered by account. We also conduct inference at the level of the source shocks using 9,999 Rademacher sign flips across the complete paths of 171 source assets. This procedure preserves the frozen account exposures, lagged loss state, common-factor controls, and fixed effects while treating the source-asset innovation paths as the primitive shock variation, following the source-level inference logic for shift-share regressions (Adão et al., 2019).

4.3 External Losses Create Forced Inventory

Figure 3 summarizes the design and the conditional estimating variation. Panel A shows the frozen-portfolio clock, the mean relative portfolio loss among accounts still at risk, and the number of accounts entering liquidation in each minute. First liquidations increase during the lead-in window as losses accumulate, before the concentrated core cascade at 21:13. Panels B and C partial out the controls and fixed effects in equation (12). Both panels display a positive relation between the current external portfolio loss and the onset of forced execution. Accounts receiving larger losses are more likely to enter liquidation during the ensuing minute and generate more forced inventory upon entry.

Table 3 reports the baseline and account fixed-effect estimates side by side and scales the external portfolio loss in units of 10 percentage points of frozen equity. Panel A examines first-liquidation entry. A 10-percentage-point external loss raises the probability of entering liquidation during the ensuing minute by 2.02 percentage points, approximately 4.1 times the unconditional mean of 0.49 percent. Adding account fixed effects leaves the estimate essentially unchanged at 1.99 percentage points.

Panel B examines initial forced inventory across the at-risk population. A 10-percentage-point external loss raises the bounded entry-minute forced-inventory rate by 1.72 percentage points, approximately 4.5 times its unconditional mean of 0.38 percent. Adding account fixed effects leaves the estimate similar at 1.68 percentage points. The close correspondence between the paired specifications shows that the external-loss relation appears both across accounts with similar frozen risk states and within the same account as its frozen portfolio receives different innovations over time.

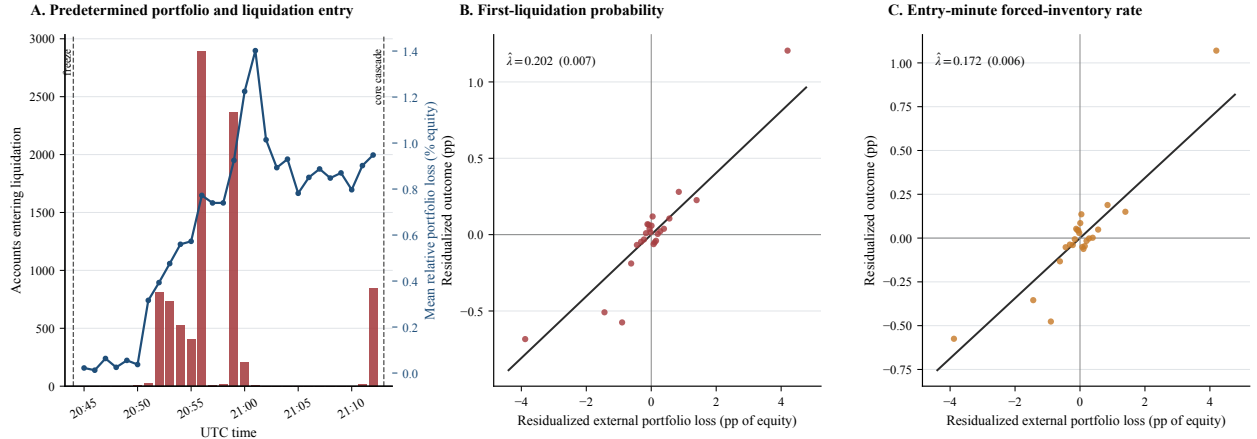


Figure 3: External Portfolio Shocks and the Onset of Liquidation

Notes: Panel A shows the frozen-portfolio clock, the mean relative portfolio loss among accounts still at risk, and the number of accounts entering liquidation in each minute. Panels B and C are residualized binned-scatter plots corresponding to the baseline specification in equation (12). Both the external portfolio loss and each outcome are residualized with respect to the lagged relative-loss state, current and lagged common-factor losses, predetermined account controls, minute fixed effects, and frozen risk-state-by-minute fixed effects. Observations are sorted into 24 equal-sized bins of residualized external loss; dots report within-bin means, and the solid lines report fitted relations estimated using the full account-minute sample. Panel coefficients are effects per one-percentage-point loss, with account-clustered standard errors in parentheses. Table 3 reports effects per ten-percentage-point loss and source-level inference.

Taken together, the two panels show that relative portfolio losses increase both liquidation entry and the total amount of initial forced inventory generated across the at-risk population. Under the one-minute no-prior-flow restriction, both point estimates fall by roughly half and source-level inference does not reject zero; under the nested two-minute restriction, both estimates exceed the benchmark and are marginally significant. Because the filters change the composition of source-minute variation, we treat them as sensitivity exercises rather than preferred specifications. For the benchmark coefficients, account-clustered inference and conditional sign randomization reject zero for both outcomes, whereas stratified complete-path permutation yields p -values of 0.105 and 0.150; Appendix C reports the distinct assignment restrictions and the complete source-level diagnostics.

These estimates provide quasi-experimental evidence on the production link in forced liquidation: asset-specific oracle innovations interacting with predetermined portfolios move fragile accounts into forced execution and generate initial forced inventory. Section 5 next holds liquidation entry fixed near the policy cutoff and estimates how the rule governing the initial release of that inventory affects execution costs, local price pressure, and the subsequent liquidation of other accounts.

Table 3: External Portfolio Losses and Initial Forced Inventory

	(1)	(2)
<i>Panel A. First-liquidation probability</i>		
External portfolio loss	2.021*** (0.072)	1.992*** (0.072)
Source-level p -value	0.0001	0.0004
Minute fixed effects	Yes	Yes
Risk-state $\times$ minute fixed effects	Yes	Yes
Account fixed effects	No	Yes
Observations	1,809,043	1,809,043
R^2	0.014	0.014
<i>Panel B. Entry-minute forced-inventory rate</i>		
External portfolio loss	1.717*** (0.065)	1.684*** (0.065)
Source-level p -value	0.0001	0.0005
Minute fixed effects	Yes	Yes
Risk-state $\times$ minute fixed effects	Yes	Yes
Account fixed effects	No	Yes
Observations	1,809,043	1,809,043
R^2	0.015	0.016

Notes: Both panels are estimated at the account-minute level in the first-liquidation risk set from 20:45 through 21:12 UTC on October 10. External portfolio loss combines frozen 20:44 account-asset exposures with asset-specific oracle innovations and is scaled to a ten-percentage-point loss relative to frozen equity. Panel A's dependent variable is an indicator for first liquidation in the minute. Panel B's forced-inventory rate is entry-minute forced-liquidation notional divided by frozen cross-margin gross notional and capped at one. Coefficients and standard errors are reported in percentage points.

Both columns include the lagged relative-loss state, current and lagged common-factor losses, pre-determined account controls, minute fixed effects, and frozen equity-, leverage-, and reconstructed-distance-group-by-minute fixed effects. Column (2) additionally absorbs account fixed effects.

The reported R^2 is the within R^2 after absorbing the listed fixed effects. Account-clustered standard errors are in parentheses. Source-level p -values use 9,999 Rademacher sign flips of the complete paths of 171 source assets. Source-score contributions imply reciprocal-Herfindahl effective source counts of 7.1 for liquidation entry and 6.7 for the forced-inventory rate; Appendix C reports source concentration and leave-source sensitivity. ***, **, and * denote significance at the 1%, 5%, and 10% levels using conditional sign-randomization source-level inference. For the benchmark estimates in column (1), stratified complete-path permutation yields p -values of 0.105 and 0.150 for Panels A and B, respectively; Appendix C describes the procedure.

5 Liquidation Policy, Execution Costs, and Subsequent Liquidation

Section 4 shows how losses on predetermined portfolios move fragile accounts into liquidation and create forced inventory. We now study what happens when that inventory reaches the market. The quantity submitted at entry is determined by the liquidation engine rather than by the distressed trader, and its immediate market effect depends on the aggressiveness of execution. Prior work shows that forced transactions can move prices when distressed sellers must trade quickly and natural liquidity providers have limited capacity (Coval and Stafford, 2007; Ellul et al., 2011; Falato et al., 2021; Lehar and Parlour, 2022). The policy can operate through both mechanical depth consumption and endogenous liquidity supply: when forced orders are predictable, strategic traders may withdraw liquidity or trade ahead of the distressed seller (Brunnermeier and Pedersen, 2005). We exploit the discontinuity in the platform’s liquidation rule at a pre-liquidation position value of \$100,000 to estimate how a policy-induced reduction in forced-sale aggressiveness changes execution costs, local price pressure, and subsequent same-asset liquidation. The discontinuity estimates the resulting total reduced-form effect of the liquidation regime rather than decomposing these channels.

5.1 The \$100,000 Threshold and Empirical Design

As described in Section 2.3, the liquidation engine first attempts to restore margin compliance by submitting an order to the ordinary limit-order book. During the sample period, a liquidatable position below \$100,000 faced an initial order for the full position. For a position above \$100,000, the engine initially targeted approximately 20% of the position and then applied a 30-second protocol interval to subsequent liquidation handling. Positions immediately around the threshold had therefore already entered the same market-method liquidation process, but the quantity initially submitted to the book changed sharply at the cutoff.

The analysis focuses on liquidated long positions, for which liquidation creates forced sale supply. We retain the first account–asset market-method liquidation episode on each UTC date, exclude events preceded by another liquidation of the same account during the prior 30 seconds, and exclude simultaneous multi-asset liquidations recorded for the same

account in the same millisecond. These restrictions isolate a single initial policy assignment and its immediate outcomes from recent or simultaneous liquidation activity. The primary bandwidth contains events with pre-liquidation position value within \$20,000 of the threshold. The resulting sample consists of 3,562 events from 2,813 accounts, 141 assets, and 200 UTC dates. Section 5.4 applies a further predetermined isolation restriction to this sample.

For liquidation event i , pre-liquidation position value is

$$N_i = |Position_{i,pre}| \times MarkPrice_{i,pre}. \quad (13)$$

We center the running variable at the policy threshold and express it in units of \$10,000:

$$R_i = \frac{N_i - 100,000}{10,000}, \quad Z_i = \mathbf{1}\{N_i > 100,000\}, \quad (14)$$

where Z_i indicates assignment to the partial-liquidation regime. Let Q_i^{first} denote the cumulative quantity filled under the first liquidation order. The realized initial-order executed share is

$$D_i = \min \left\{ \frac{Q_i^{first}}{|Position_{i,pre}|}, 1 \right\}. \quad (15)$$

We measure execution and price effects at the event level. Let $VWAP_i$ be the volume-weighted execution price of the initial liquidation order. Because every event in the primary sample is a forced sale, adverse execution slippage relative to the pre-liquidation mark is

$$Slippage_i = - \left(\frac{VWAP_i}{MarkPrice_{i,pre}} - 1 \right) 10,000. \quad (16)$$

Positive values indicate that the position was sold below the mark. Let $P_{i,pre}$ denote the last valid transaction price before liquidation and $P_{i,h}$ the transaction price at horizon h . Directional transaction-price stress is

$$PriceStress_{i,h} = - \log \left(\frac{P_{i,h}}{P_{i,pre}} \right) 10,000. \quad (17)$$

We report transaction-price stress at ten seconds and the maximum adverse transaction-price

movement during the first ten seconds,

$$MaxAdverseMove_{i,[0,10]} = \max_{s \in [0,10]} \left\{ -\log \left(\frac{P_{i,s}}{P_{i,pre}} \right) 10,000 \right\}. \quad (18)$$

5.2 Local-Linear Specification and Validity

For outcome Y_i , we estimate the reduced-form discontinuity

$$Y_i = \alpha + \tau Z_i + \beta_- R_i(1 - Z_i) + \beta_+ R_i Z_i + \mu_{a(i)} + \delta_{d(i)} + \eta_{h(i)} + u_i. \quad (19)$$

Here, Y_i denotes an event-level outcome: the initial-order executed share, execution slippage, short-horizon directional transaction-price stress, or the number and initial-order notional of subsequent liquidation onsets. The running variable R_i is the pre-liquidation position value centered at \$100,000 and scaled by \$10,000, and Z_i indicates assignment to the partial-liquidation regime. The coefficients β_- and β_+ are the local slopes below and above the threshold, respectively. The terms $\mu_{a(i)}$, $\delta_{d(i)}$, and $\eta_{h(i)}$ denote asset, UTC-date, and UTC-hour fixed effects, and u_i is the event-level residual. The coefficient of interest, τ , captures the discontinuous change in Y_i at the threshold and measures the local effect of assignment to the partial-liquidation regime among positions already entering market-method liquidation. Because the threshold changes both the initial-order target and the subsequent 30-second handling path, the primary estimand is the reduced-form effect of assignment to the partial-liquidation regime. The discontinuity in D_i documents the associated change in initial execution intensity.

The specification assigns triangular-kernel weights to observations within \$20,000 of the cutoff, following standard local-linear regression-discontinuity practice (Imbens and Lemieux, 2008). Standard errors for the initial-order executed share and execution slippage are two-way clustered by account and UTC date; price and subsequent-liquidation outcome standard errors are two-way clustered by asset and UTC date. Internet Appendix D supplements the fixed-bandwidth estimates with robust bias-corrected inference and MSE-optimal bandwidths following Calonico et al. (2014).

The running-variable density for long positions is smooth at the cutoff; the local-polynomial density test of Cattaneo et al. (2020) yields a p -value of 0.315. Pre-event oracle returns, prior market-wide forced flow, the mid-to-oracle basis, oracle volatility, account

equity, and cross-margin leverage also vary smoothly through the threshold. Placebo cutoffs at \$90,000 and \$110,000 produce no discontinuity in the initial-order executed share. Internet Appendix D reports the density and balance estimates, alternative bandwidths, a donut specification, the sample excluding Bitcoin, and the placebo-cutoff results.

5.3 Execution Costs and Local Price Effects

Figure 4 presents the threshold evidence. Panel A shows the institutional first stage. Immediately below the cutoff, a large share of the position is executed under the initial liquidation order; immediately above it, the corresponding share is approximately 20%. Assignment to the partial-liquidation regime lowers the initial-order executed share by 50.0 percentage points. The first-stage F -statistic is 648.3.

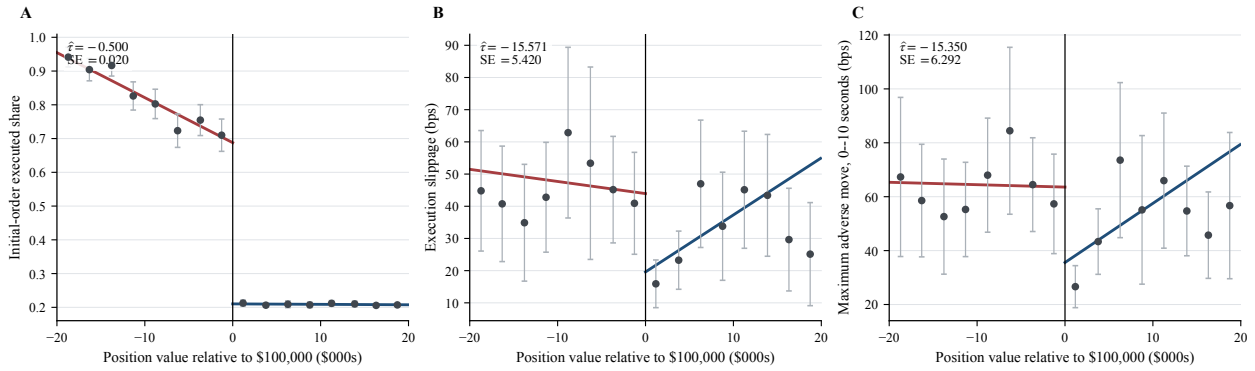


Figure 4: The \$100,000 Policy Threshold and Local Execution Effects

Notes: The sample consists of long positions already entering market-method liquidation with pre-liquidation position value within \$20,000 of the \$100,000 policy cutoff. The vertical line marks the cutoff. Dots report \$2,500-bin means with 95% confidence intervals, and solid lines are side-specific weighted local fits to the unadjusted outcomes. Panel annotations report the corresponding discontinuities and clustered standard errors from equation (19), which includes separate local slopes and asset, UTC-date, and UTC-hour fixed effects. Panel A is reported in percentage points; Panels B and C are reported in basis points.

Panels B and C show that the change in the initial-order executed share is accompanied by discrete improvements in both execution quality and the immediate price path. Crossing into the partial-liquidation regime lowers adverse execution slippage by 15.571 basis points. The maximum adverse transaction-price movement during the first ten seconds falls by 15.350 basis points. Both discontinuities occur at the same threshold as the change in the initial-

order executed share.

Panel A of Table 4 reports the immediate local-linear estimates. Directional transaction-price stress measured exactly ten seconds after liquidation falls by 8.267 basis points and is less precisely estimated than the execution-slippage and maximum-adverse-move effects. The full set of immediate outcomes points in the same direction: assignment to the less aggressive regime improves the price received by the liquidated account and attenuates the downside transaction-price path immediately following the forced sale.

The magnitudes are economically meaningful relative to the sample means reported in Table 1. Mean adverse execution slippage is 39.4 basis points, so the 15.6-basis-point discontinuity represents approximately 40% of the sample mean. The mean maximum adverse movement during the first ten seconds is 58.1 basis points; the estimated policy effect is approximately 26% of that mean. Execution policy therefore governs how much of a given quantity of distressed inventory is converted into contemporaneous trading costs and local price dislocation.

5.4 Feedback into Subsequent Liquidation

We next apply the same specification to an isolation-restricted sample. Within the same asset and bandwidth, a focal event for horizon h is retained only when no other eligible focal event occurred during the preceding h seconds. This restriction uses only information observed before the focal event and therefore does not select the sample on subsequent outcomes. To compare the 30-, 60-, and 300-second horizons on a common sample, Panel B of Table 4 applies the strict 300-second isolation rule and requires complete 300-second support, yielding 2,067 focal events.

A new liquidation onset is the first market-liquidation burst for another account after more than 30 seconds without a forced fill through either the market or backstop route. We exclude the focal account, count each other account once within each outcome window, and sum the initial liquidation-order notional generated at those onsets. Equation (19) is estimated without modification on this isolation-restricted sample. The institutional first stage remains sharp: assignment to the partial-liquidation regime lowers the initial-order executed share by 0.461, with a partial F -statistic of 84.5. The density test does not reject continuity in this sample ($p = 0.701$), and the corresponding pre-event notional and account-

count placebos are insignificant at all three horizons.

Panel B reports the subsequent-liquidation estimates. Assignment to the less aggressive regime reduces new liquidation notional by \$0.111 million within 30 seconds, \$0.181 million within 60 seconds, and \$0.357 million within 300 seconds. The corresponding account estimates are -3.650 , -6.507 , and -22.386 . The estimates show that liquidation design affects not only the focal account’s execution but also the breadth of new same-asset liquidation that follows, with complementary evidence on notional. Across robustness specifications, the account-count estimate remains negative, whereas the notional estimate is more sensitive to bandwidth and sample exclusions; we therefore emphasize the account count and treat notional as complementary. Robust bias-corrected estimates with data-driven bandwidths preserve the discontinuities in initial execution and the direction of the local price effects. The downstream account-count estimate remains negative but is less precise, while downstream notional is sensitive to bandwidth and covariate adjustment; Internet Appendix D reports both unadjusted and indicator-adjusted estimates.

Table 4: The Partial-Liquidation Policy: Execution and Subsequent Liquidation

Panel A. Initial Forced Execution and Immediate Local Price Effects

	First stage	Reduced-form outcomes		
Dep. var.:	Initial-order executed share (pp) (1)	Execution slippage (bps) (2)	Transaction-price stress at 10 sec. (bps) (3)	Max. adverse move 0–10 sec. (bps) (4)
Above \$100,000	-49.998*** (1.964)	-15.571*** (5.420)	-8.267* (4.642)	-15.350** (6.292)
Asset FE	Yes	Yes	Yes	Yes
UTC-date FE	Yes	Yes	Yes	Yes
UTC-hour FE	Yes	Yes	Yes	Yes
Bandwidth	\$20,000	\$20,000	\$20,000	\$20,000
Observations	3,562	3,562	3,490	3,524
First-stage <i>F</i> -statistic	648.3			

Panel B. Subsequent New Liquidation Onsets

	New liquidation notional (\$ millions)			Newly liquidated accounts		
Horizon:	30 sec. (1)	60 sec. (2)	300 sec. (3)	30 sec. (4)	60 sec. (5)	300 sec. (6)
Above \$100,000	-0.111** (0.052)	-0.181* (0.097)	-0.357** (0.140)	-3.650 (3.836)	-6.507 (4.513)	-22.386*** (7.745)
Asset FE	Yes	Yes	Yes	Yes	Yes	Yes
UTC-date FE	Yes	Yes	Yes	Yes	Yes	Yes
UTC-hour FE	Yes	Yes	Yes	Yes	Yes	Yes
Bandwidth	\$20,000	\$20,000	\$20,000	\$20,000	\$20,000	\$20,000
Observations	2,067	2,067	2,067	2,067	2,067	2,067
First-stage <i>F</i> -statistic			84.5			

Notes: Panel A retains the original sample of long positions already entering market-method liquidation with pre-liquidation position value within \$20,000 of the \$100,000 policy cutoff. Panel B additionally requires complete 300-second support and no earlier eligible focal event in the same asset during the preceding 300 seconds; all six columns therefore use the same 2,067 focal events. Above \$100,000 indicates assignment to the less aggressive partial-liquidation regime.

Each column reports the discontinuity from a triangular-kernel local-linear regression with separate slopes and asset, UTC-date, and UTC-hour fixed effects. Panel B counts, for the same asset, the first new market-liquidation onset of every other account during the stated horizon. New liquidation notional is the sum of those accounts' initial liquidation-order notionals, not total subsequent fills. Negative Panel B estimates indicate fewer subsequent liquidation onsets under the less aggressive regime. We treat the number of newly liquidated accounts as the primary downstream outcome; new-liquidation notional is complementary because it is more sensitive to bandwidth and covariate adjustment.

The initial-order executed share is reported in percentage points; slippage and price outcomes are reported in basis points. Standard errors for Panel A's initial-order share and slippage are two-way clustered by account and UTC date; Panel A price outcomes and all Panel B outcomes are two-way clustered by asset and UTC date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

The downstream account-count effect is also economically material. At five minutes, 22.4 fewer newly liquidated accounts represents 27.8% of the common sample mean of 80.5 accounts. The complementary notional estimates equal 41.4% of the 30-second mean and 33.4% of the five-minute mean. In the fixed-bandwidth common-sample estimates, the notional coefficient appears within 30 seconds, while the account-count coefficient is largest over five minutes. These horizons summarize cumulative downstream outcomes rather than a formal decomposition of their timing.

5.5 State Dependence and the October 10 Bridge

The largest local price effects occur in high-stress observations, especially during the October 10 episode. We classify an event as high stress when a pre-liquidation index combining the absolute five-minute oracle return and aggregate market-method forced-liquidation flow during the preceding 60 seconds lies in its top quintile. In an interaction specification, the additional ten-second price discontinuity in high-stress states is -25.924 basis points. This concentration connects the full-sample threshold evidence to the extreme-state anatomy below. Internet Appendix D reports the interaction estimates, the October 10 exclusion sensitivity, and the short-horizon price path.

Together, Sections 4 and 5 identify three consecutive mappings. Losses generated by predetermined portfolio exposures create forced inventory; the policy governing its initial release determines immediate execution costs and local price pressure; and that release also determines how much additional forced liquidation appears among other accounts in the same asset. The next section provides high-resolution accounting of how the relevant objects appear together during October 10, tracing valuation-path exposure, initial recipients, and downstream ADL.

6 Inside the October 10 Liquidation Cascade: High-Resolution Event Accounting

Sections 4 and 5 identify three consecutive links in forced liquidation. Losses on predetermined portfolios create forced inventory; the liquidation policy governing its release changes immediate trading costs and local transaction prices; and the release rule also determines how much subsequent liquidation appears among other accounts in the same asset. This section places the three identified links inside the full October 10 cascade and provides high-resolution accounting of valuation-path exposure, subsequent liquidation, initial receipt, and downstream ADL. Across venues, the episode produced more than \$19 billion in liquidations, making it the largest recorded cryptocurrency liquidation event.⁹ The event reconstruction follows the system from the pre-burst price decline through concentrated forced flow, additional balance-sheet exposure along the realized valuation path, subsequent liquidation, initial receipt, and downstream ADL. This high-resolution anatomy follows the audit-trail event-reconstruction tradition of Kirilenko et al. (2017) and complements event reconstructions of runs and collateral-liquidation feedback in digital-asset markets (Liu et al., 2023; Lehar and Parlour, 2022). The account-to-market data show how the objects measured locally coexist when many constrained balance sheets reach the market at once.

6.1 Event Clock and Local Market Dislocation

We organize the event around a common minute-level clock. The account state used in Section 4 is frozen at 20:44 UTC. The lead-in extends through 21:12, the core liquidation burst runs from 21:13 through 21:20, and the early resolution period begins at 21:21. Figure 5 aligns perpetual prices, the mid-to-oracle basis, impact-price conditions, displayed depth, gross forced-side liquidation, and ADL over this sequence.

Panel A indexes the Hyperliquid perpetual mid prices of BTC, ETH, and SOL to 100 at the beginning of the clock. Their troughs reach 89.4, 82.8, and 68.3, respectively. The decline becomes markedly steeper after the core burst begins, particularly for SOL. Panel B shows the corresponding deterioration in execution conditions across the asset universe.

⁹The cross-venue total is based on CoinGlass aggregation reported by CoinDesk Research on October 17, 2025. All Hyperliquid quantities in this paper are reconstructed from the underlying platform records.

The open-interest-weighted mid-to-oracle basis reaches -426.9 basis points, while the impact gap reaches 340.4 basis points. The negative basis indicates that local perpetual mids trade below the underlying spot-oracle anchor; the widening impact gap indicates a sharp increase in the cost of immediately demanding liquidity.

Panel C shows that aggregate displayed depth within five basis points contracts as the market enters the liquidation burst. Bid depth measures the displayed capacity to absorb forced sales of long positions, whereas ask depth absorbs forced purchases of short positions. Because the panel aggregates assets with different liquidation directions, the two series need not move symmetrically or track aggregate forced flow one-for-one. Backstop transfers do not themselves consume displayed order-book depth.

Panel D records the arrival of forced inventory. Gross forced-side flow reaches \$814.4 million in the first core-burst minute. It then peaks at \$1.583 billion at 21:17 and \$1.466 billion at 21:19. Backstop-method rows supply most of the flow at both peaks. ADL appears in the same concentrated interval, reaching \$765.8 million and \$738.3 million at 21:17 and 21:19. On the phase-aligned adjustment clock reported in Internet Appendix E, high-fragility accounts privately reduce \$675.1 million during the 29-minute lead-in, while \$601.8 million of forced liquidation also occurs. During the eight-minute core burst, forced liquidation rises to \$5.643 billion, compared with \$439.4 million of continued private reduction. Private adjustment is therefore visible before the burst, but forced inventory dominates once the cascade reaches its core.

The timing connects the two earlier designs to the event anatomy. External portfolio losses generate liquidation entry during the lead-in, while the threshold estimates show that aggressive forced execution worsens transaction prices most strongly in high-stress states and increases subsequent liquidation by other accounts. October 10 combines these elements: a dense distribution of fragile accounts, rapidly arriving forced inventory, a sharply negative perpetual basis, and limited contemporaneous depth.

6.2 Valuation-Path Exposure and Subsequent Liquidation

We next measure how the realized mark path maps into account margin states. The realized scenario uses the minute-level mark price, which is the price entering unrealized profit and loss, maintenance calculations, and liquidation decisions. The benchmark uses the official

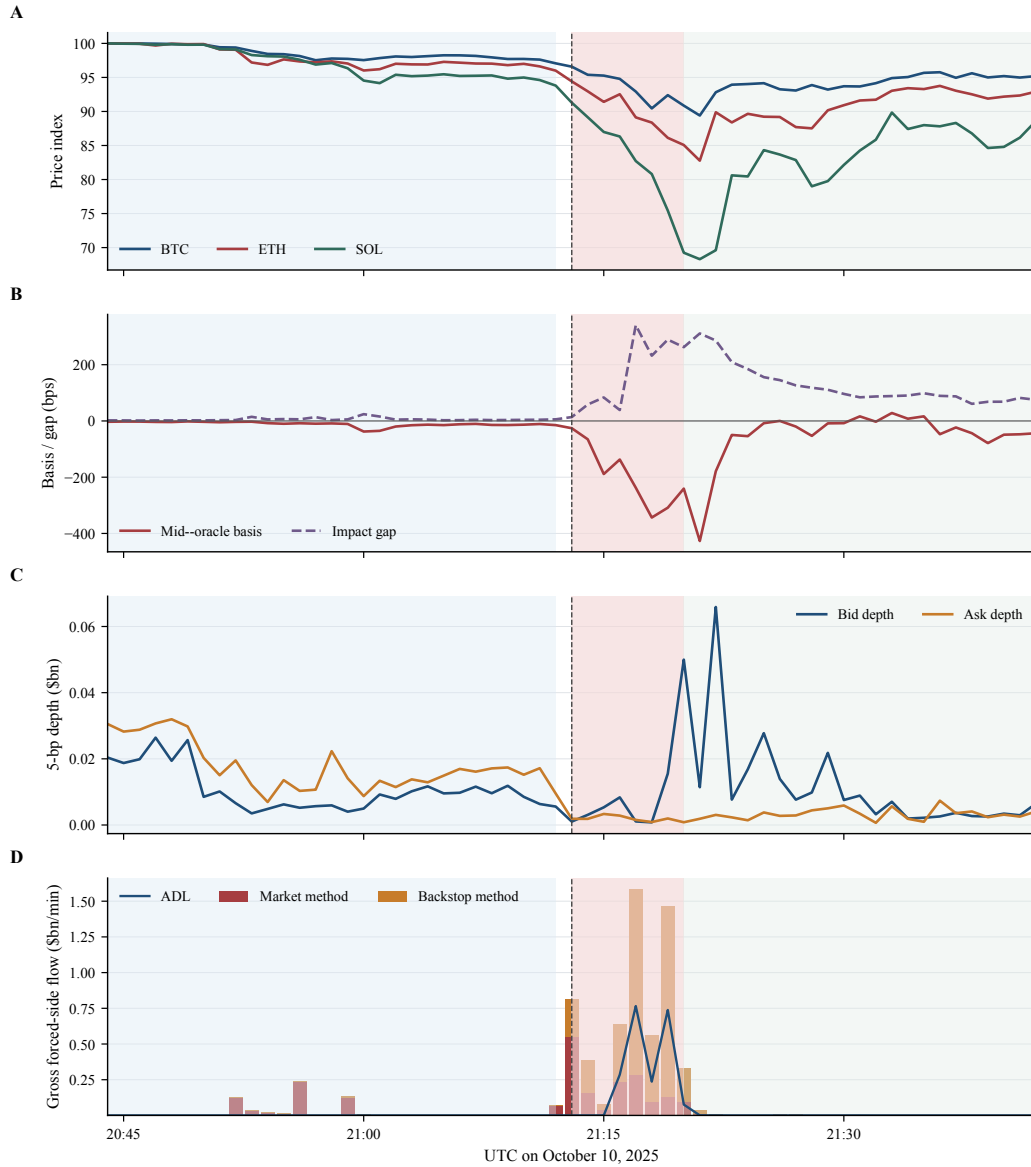


Figure 5: October 10 Event Clock and Market Dislocation

Notes: The figure covers 20:44–21:42 UTC on October 10, 2025. Blue, red, and green shading mark the lead-in (20:44–21:12), core liquidation burst (21:13–21:20), and early resolution (21:21–21:42); the dashed vertical line marks the start of the core burst. Panel A indexes the Hyperliquid perpetual mid prices of BTC, ETH, and SOL to 100 at 20:44. Panel B reports open-interest-weighted averages across assets; the mid-to-oracle basis is measured relative to the official spot oracle, and impact bid and ask are book-implied execution prices at the platform’s applicable impact notional, and the impact gap is the impact ask minus the impact bid, scaled by the mid. Panel C sums, across 182 assets, the dollar value of displayed bid and ask liquidity within five basis points of each asset’s mid. These are raw side-specific aggregates rather than pressure-aligned measures, and backstop transfers do not mechanically consume this displayed depth. Panel D stacks forced-side liquidation fills by the protocol’s market and backstop methods. ADL plots the selected-counterparty side of matched ADL trades and is not added to the forced-side bars.

spot oracle. Because the oracle already contains the common decline in the underlying market, the comparison asks how much additional balance-sheet exposure is associated with the realized mark path relative to the official spot-oracle benchmark.

At each minute t , we reconstruct the positions held by account i across assets a . Here, i , a , and t index accounts, assets, and minutes, respectively. Within each account-minute, we hold the observed position vector fixed and value the same portfolio twice: first using the realized mark prices and then using the official spot-oracle prices. Positions may evolve across minutes as observed, but the two scenarios use identical positions within a given account-minute. This procedure produces two minute-by-minute margin-state paths for every account.

Let $s \in \{M, O\}$ index the valuation scenario, where M denotes the realized mark path and O denotes the official spot-oracle benchmark. Let A_{it} be account i 's observed account equity (the platform's account-value field) at minute t , q_{iat} its signed position quantity in asset a , P_{at}^M the realized mark price, P_{at}^s the price used under scenario s , and MM_{iat} the position's observed maintenance-margin contribution. The resulting maintenance surplus of account i at minute t under scenario s is

$$B_{it}^s = A_{it} + \sum_a q_{iat} (P_{at}^s - P_{at}^M) - \sum_a MM_{iat} \frac{|q_{iat} P_{at}^s|}{|q_{iat} P_{at}^M|}. \quad (20)$$

The first term is account equity observed under the realized mark. The second term revalues the same within-minute positions from the mark to the scenario price, while the final term adjusts the maintenance requirement for the corresponding change in position notional. When $s = M$, the expression reproduces the account's realized margin state. When $s = O$, it measures the margin state of the same observed portfolio valued at the official oracle. Repeating this calculation for every account and minute yields the two account-level margin paths used below.

An account is in margin shortfall when $B_{it}^s < 0$. For each scenario, we count the account once at its first observed shortfall during the core burst and measure the gross position notional held at that observation, $\sum_a |q_{iat} P_{at}^s|$. The resulting dollar amount is therefore shortfall-exposed position notional: a first-observed balance-sheet stock rather than the dollar deficit below maintenance or cumulative liquidation flow. It is not expected to reconcile to the \$6.666 billion of event-wide gross forced-transfer notional reported below.

Panels A and B of Figure 6 report the cumulative paths of first observed shortfalls during the core burst. Panel A counts each account once at its first observed margin shortfall during the core burst, while Panel B cumulates the gross position notional held at those first observations. The two valuation paths remain close at the start of the burst and then separate sharply as the local mark dislocation deepens. Under the realized mark valuation, an additional \$161.9 million of gross position notional is in margin shortfall relative to the official spot-oracle benchmark. Table 5 reports the corresponding account counts and scenario-level stocks. The mark is institutionally anchored to the oracle, so this comparison removes the much larger common spot decline. Its economic significance comes from the distribution of margin buffers: a moderate additional valuation wedge can move many already fragile balance sheets across the maintenance boundary.

We then connect path-dependent exposure to subsequently realized forced flow. We label an account as at risk when its maintenance-surplus distance is no greater than 5%. The first wave contains accounts that are at risk or in margin shortfall under the oracle benchmark during the lead-in or core burst. The second wave contains accounts that become at risk under the realized mark path during the core burst or early resolution period but never enter the corresponding risk set under the oracle benchmark. Each account is labeled at its first qualifying observation, after which we sum its realized forced-liquidation flow. Thus, the first wave captures fragility already implied by the common oracle-price decline, whereas the second captures incremental fragility that appears only along the realized mark path; the distinction is path-based rather than merely chronological.

Panel C and Table 5 connect these labeled balance sheets to subsequently realized forced flow. Accounts classified into the second wave subsequently account for \$242.6 million in forced liquidation, equal to 11.4% of the flow attributed to the oracle-exposed first wave. Panel C traces the accumulation of both waves' realized flows from each account's classification minute onward, while Table 5 reports their account counts and first-observed at-risk and shortfall-exposed balance-sheet stocks.

The comparison supplies a stock-to-flow link inside the event. The oracle benchmark classifies balance sheets exposed by the common underlying decline; the realized mark path classifies an additional set exposed along the observed perpetual-market valuation path; and the account links show that both groups later produce forced inventory. Section 5 identifies both the local price effect of forced-sale aggressiveness and its effect on subsequent liquidation

by other accounts. The path comparison here measures the event-scale counterpart of that second effect: how many balance sheets the realized valuation path placed at risk, and how much forced liquidation they subsequently produced. The two objects are not the same estimand: Section 5 varies execution aggressiveness at a single position threshold within an asset, whereas the path comparison varies the valuation benchmark for the entire event.

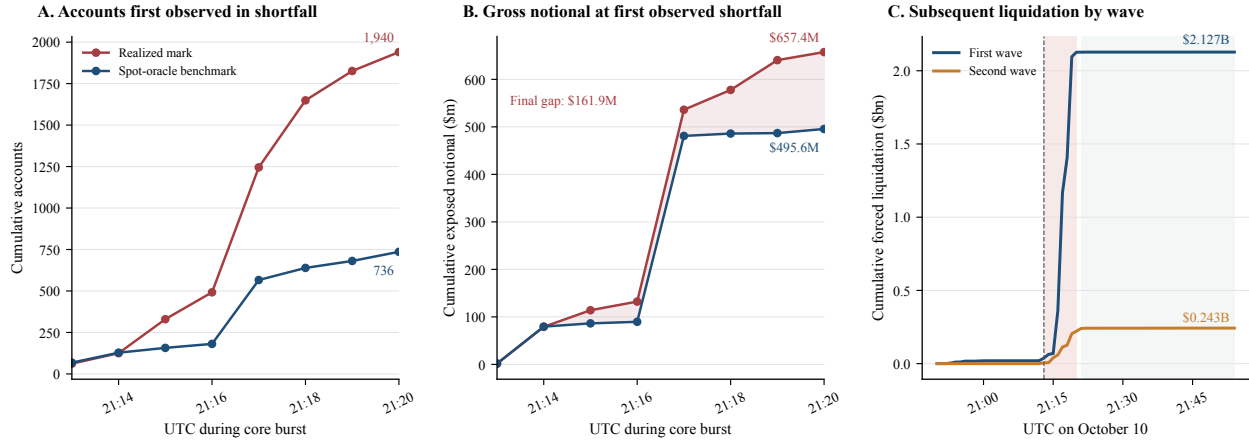


Figure 6: Valuation-Path Exposure and Subsequent Liquidation

Notes: Panels A and B count each account once at its first observed margin shortfall during the core burst. Panel A cumulates accounts, while Panel B cumulates gross position notional measured at each account's first shortfall observation. The realized path uses the mark price and the benchmark uses the official spot oracle. Panels A and B therefore report first-observed balance-sheet stocks. Panel C instead cumulates subsequently realized forced-liquidation fills for the predefined waves and reports a flow; these quantities are not additive and are not expected to reconcile.

6.3 Initial Receipt and Downstream ADL

The final part of the event reconstruction follows how risk is transferred after the initial liquidation. When risk-bearing capital is specialized and balance-sheet constrained, prices and allocations depend on which balance sheets can enter and remain in the market (Shleifer and Vishny, 1997; Gromb and Vayanos, 2002; Duffie, 2010). Figure 7 follows the sequence from initial forced liquidation, to the initial recipient of the position, and then to downstream ADL. Figure 5 establishes when these mechanisms operate; Figure 7 summarizes their event-wide scale and concentration.

Panel A begins with \$4.667 billion of initial forced liquidation sourced by accounts outside the identified HLP/system set. It partitions the same initial flow once by the initial

liquidation method applied to the forced account and again by the identity of the initial recipient; these are alternative reconciliations and are not additive. Approximately half of the initial flow is received initially by identified HLP/system accounts. Table 5 reports the complete liquidation-method and recipient decomposition.

Measured against the subsequent realized perpetual price, matched positions received by identified HLP/system accounts have an adverse raw aggregate markout of \$76.4 million over five minutes, before the markout turns positive over thirty minutes (Internet Appendix Figure G1). This raw path measures the market exposure placed on recipient balance sheets, while an oracle-relative markout measures the local execution concession.

Using a six-hour benchmark and tracking the subsequent vault unwind, Jia et al. (2026) report positive longer-horizon backstop markouts and approximately \$50 million in net vault profit; our one- to thirty-minute markouts instead measure the short-horizon exposure borne at initial receipt, before subsequent unwinding.

The accounting next records a downstream forced layer. Identified HLP/system accounts later appear as the forced-side accounts in \$1.999 billion of ADL-linked trades. Combined with the \$4.667 billion initial flow, gross forced-transfer notional across the two observed layers is \$6.666 billion. At the account-set level, identified HLP/system accounts therefore appear in both initial receipt and downstream forced reallocation. Studies of ADL design use the same event (Chitra, 2025; Chitra et al., 2026).

Panel B shows that initial identified HLP/system receipt is highly concentrated: the two largest accounts receive 99.46% of the identified amount. During October 10, this concentration, together with the subsequent forced flow sourced by identified HLP/system accounts, places initial receipt on a narrow set of institutional balance sheets.

Table 5 closes the accounting. Panel A compares shortfall-exposed balance-sheet stocks across valuation paths, Panel B links the predefined waves to later realized liquidation, and Panel C reconciles initial liquidation methods, initial recipients, and downstream ADL. During October 10, the forced-liquidation sequence passes through public liquidity, concentrated backstop balance sheets, and involuntary deleveraging.

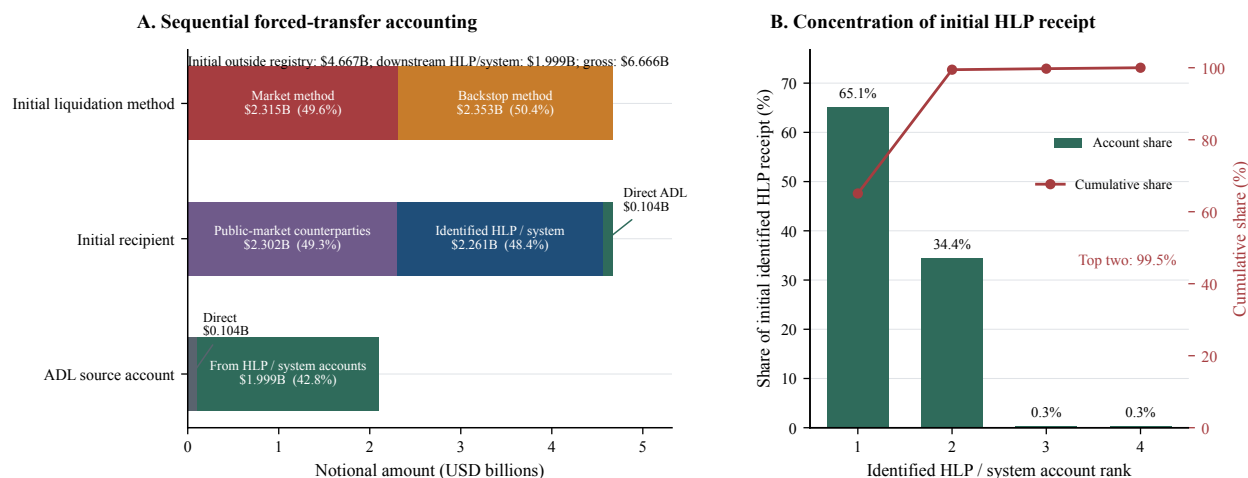


Figure 7: Forced-Transfer Accounting and Concentration of Initial Institutional Receipt

Notes: Panel A should be read as three separate accounting blocks. The first block partitions \$4.667 billion of initial forced liquidation sourced by accounts outside the identified HLP/system set according to the initial liquidation method. The second partitions the same \$4.667 billion according to the initial recipient. Each block sums to \$4.667 billion, so the two blocks must not be added together. The third block partitions \$2.103 billion of ADL according to the identity of the forced-side account. Direct ADL from accounts outside the identified set is already part of the initial \$4.667 billion; the \$1.999 billion sourced subsequently by HLP/system accounts is the second layer included in the \$6.666 billion gross forced-transfer total. Adding total ADL to that gross total would therefore double count the same matched trades. Percentages use the initial \$4.667 billion as their denominator. ADL-selected counterparties are not classified as voluntary private recipients. Panel B ranks four identified HLP/system accounts by initial, non-ADL-linked receipt; it does not report terminal holdings.

Table 5: October 10 Valuation-Path Exposure and Forced-Transfer Accounting

Panel A. Additional balance-sheet exposure in the core burst					
Valuation path		Accounts	Gross position notional in shortfall (\$m)		
Realized mark path		1,940	657.417		
Official spot-oracle benchmark		736	495.550		
Realized minus oracle benchmark			161.867		
Panel B. Realized liquidation waves (first-wave flow = 100)					
Wave	Accounts	Gross position notional (\$m)		Subsequent forced liquidation (\$m)	Share of first-wave flow (%)
		At risk	In shortfall		
First wave	1,741	547.512	501.286	2,127.417	100.00
Second wave	1,830	278.387	186.061	242.638	11.41
Panel C. Initial Receipt and Downstream ADL					
Accounting component			Amount (\$m)	Share of initial flow (%)	
Initial forced liquidation sourced by accounts outside the identified HLP/system set			4,667.158	100.00	
Gross forced-transfer notional			6,666.446		
Initial liquidation method					
Market method			2,314.570	49.59	
Backstop method			2,352.588	50.41	
Initial recipient					
Public-market counterparties			2,302.209	49.33	
Identified HLP/system accounts			2,260.890	48.44	
ADL-selected counterparties			104.059	2.23	
ADL source					
Directly from accounts outside the identified set			104.059	2.23	
From HLP/system accounts			1,999.288	42.84	
Total ADL redistribution			2,103.347	45.07	

Notes: Panel A values the same observed account positions under the realized mark path and the official spot-oracle benchmark. It counts each account once at its first core-burst observation with negative maintenance surplus and reports the gross position notional held at that observation. These amounts are balance-sheet stocks, not maintenance deficits or liquidation flows. Because the two valuation paths need not place exactly the same accounts in shortfall, realized minus oracle is the net difference between the two scenario-level stocks.

Panel B defines an account as at risk when its maintenance-surplus distance is no greater than 5%; shortfall means that maintenance surplus is negative. The first wave is classified under the oracle benchmark during the lead-in or core burst. The second wave is classified under the realized mark path during the core burst or early resolution and excludes accounts that ever enter the corresponding oracle risk set. At-risk and shortfall notionals are measured once, at each account's first qualifying observation. Amount instead sums all subsequently realized forced-liquidation fills for the labeled accounts and is therefore a flow rather than a stock. The share column expresses each wave's flow relative to first-wave flow.

Panel C should be read as three accounting blocks. The initial-liquidation-method and initial-recipient blocks separately partition the same \$4.667 billion of initial forced liquidation sourced by accounts outside the identified HLP/system set; each block sums to that amount and the two blocks are not additive. The ADL-source block partitions \$2.103 billion by the identity of the forced-side account: \$104.1 million comes directly from accounts outside the identified set and \$1.999 billion comes subsequently from identified HLP/system accounts. Gross forced-transfer notional is \$6.666 billion, equal to the initial \$4.667 billion plus the subsequent \$1.999 billion. Total ADL is the matched side of those forced transfers and must not be added again. All Panel C shares use the initial \$4.667 billion as the denominator. Amounts are not P&L or terminal holdings; they are gross transfer notionals. Displayed amounts are rounded to the nearest \$0.001 million, and shares are calculated from unrounded values.

7 Conclusion

Forced liquidation is not a terminal transaction. This paper follows the process from portfolio loss, through forced execution, to downstream risk transfer. Two quasi-experiments identify the production and execution links: externally induced losses on predetermined portfolios increase liquidation entry and initial forced inventory, while assignment to the partial-liquidation regime changes immediate execution costs and local transaction-price stress. The same discontinuity provides additional evidence on the subsequent liquidation of other accounts. The high-resolution October 10 reconstruction then documents additional shortfall exposure along the realized mark path and accounts for subsequent forced-transfer layers. Accounts in the same identified institutional set later appear on the forced side of ADL-linked trades.

Three broader implications follow. First, fragility is distributional: aggregate leverage is an incomplete measure because the quantity of forced inventory generated by a price move depends on the density of accounts near their maintenance boundaries. Second, the market effect of forced inventory depends on the liquidation rule. How quickly that inventory reaches the market determines immediate trading costs, local price pressure, and how much additional forced liquidation follows among other accounts. Third, liquidation closes a distressed account's position without eliminating the inventory or risk from the financial system. The identity of the initial recipient and the accounts involved in downstream loss sharing are therefore part of the observed resolution process. Together, these results direct attention to the distribution of margin buffers, execution policy, and recipient balance sheets.

The estimates identify one side of a broader policy trade-off: assignment to the less aggressive partial-liquidation regime reduces the liquidated account's execution costs and adverse local price movement, with additional evidence of fewer other accounts subsequently forced into liquidation. The last outcome is an externality rather than a transfer between the distressed account and its counterparties, so the choice of liquidation rule can have consequences that a venue's own account-resolution objective need not internalize.

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Internet Appendix

Inside Forced Liquidation:

Margin Constraints, Execution, and Risk Transfer

This Internet Appendix provides measurement detail, complete specifications, and targeted robustness for the analyses in the main text. Appendix A documents the linked data and account-state reconstruction. Appendix B reports the complete broad-panel evidence. Appendices C and D document the shock-exposure and policy-discontinuity designs for forced-inventory production and local execution effects. Appendix E aligns private and forced adjustment on the October 10 event clock, and Appendix F reconciles the event’s balance-sheet stocks with transaction-level forced flows. Appendix G reports short-horizon markouts on initial receipt by identified HLP/system accounts.

A Institutional Setting, Data Construction, and Measurement

This appendix supports Figure 1 and Table 1 of the main text. It links account margin states and positions to the official spot oracle, mark prices, orders, fills, counterparties, liquidation methods, and ADL roles. A cross-margin account is the relevant balance-sheet unit because all of its cross positions share collateral. The official spot oracle measures the external price path, the mark values positions for margin purposes, and transaction and order-book prices measure local execution.

Figure A1 summarizes the linked-data architecture and flow classification. For each matched trade, the initial recipient is the account on the opposite side of the forced fill; identified HLP/system recipients are classified using the versioned account registry. The forced-side account, liquidation method, initial recipient, and ADL role remain separate objects. Table A1 reports the data universe and observation units, and Table A2 documents construction of the three broad-panel regression samples.

The October 10 account replay is anchored to the full-account snapshot at 20:04:54 UTC and advances positions and balances through timestamped fills, funding payments, and other ledger events. For eligible cross-margin accounts, the applicable maintenance-margin schedules allow position-level maintenance requirements to be aggregated at the account level. This construction supports the maintenance-distance measure used in Figure 2 of the main text.

A.1 Maintenance-Tier Reconstruction

For position j in account i at minute t , let $N_{ijt} = |q_{ijt}|P_{jt}^M$ denote absolute position notional valued at the contemporaneous mark price. The platform assigns each asset either a flat maintenance rule or an asset-specific tier schedule based on N_{ijt} . Under a tiered schedule, position notional selects tier τ , and the reconstructed position-level requirement is

$$MM_{ijt} = N_{ijt}m_{j\tau} - d_{j\tau}, \quad (\text{A.1})$$

where $m_{j\tau}$ is the maintenance-margin rate and $d_{j\tau}$ is the tier-specific deduction that keeps the requirement continuous at tier boundaries. For assets governed by a flat maximum-leverage

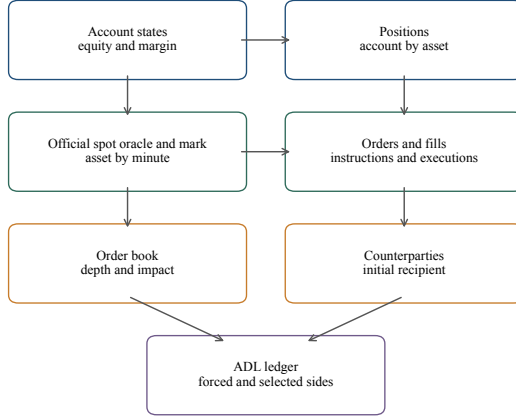
rule, the corresponding requirement is

$$MM_{ijt} = \frac{N_{ijt}}{2L_j^{\max}}, \quad (\text{A.2})$$

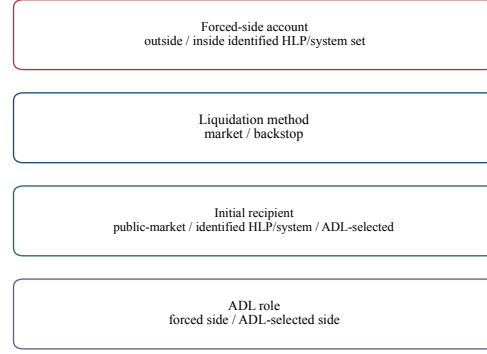
where $L_j^{\max}$ is the asset’s maximum permitted leverage. The aggregate cross-margin maintenance requirement is the sum of MM_{ijt} over all open cross positions in the account. We classify an account state as having a fully reconstructed cross-margin requirement only when every open cross position can be matched to an official flat rule or complete tier schedule in the official risk-parameter inputs. The account-level maintenance surplus and reconstructed distance to the maintenance boundary in the main text are constructed from this aggregate requirement.

The three broad-panel samples begin from related but separately organized canonical panels. All three samples are restricted to ordinary user-controlled accounts; protocol or system accounts, HLP-related accounts, and vault accounts are excluded. The private-adjustment sample retains initial account–asset positions for ordinary user-controlled cross-margin accounts with at least \$100 of equity, positive cross notional, finite positive leverage, matched endpoint benchmark prices, and a valid within-window position path. The forced-liquidation sample aggregates the same underlying initial portfolios to the account–window level and additionally requires benchmark coverage for at least 80 percent of initial cross notional and complete loss and liquidation variables. The linked-portfolio sample returns to the account–target-asset–window level, requires at least one other initial position, and retains target rows for which at least 90 percent of non-target source notional is benchmark matched. Table A2 reports every sequential restriction and the remaining observations.

A. Linked data architecture



B. Flow classification



Forced-side account, liquidation method, initial recipient, and ADL role are recorded separately.

Figure A1: Data Linkage and Flow Classification

Notes: The figure summarizes the package-local links among account states, positions, prices, instructions, fills, counterparties, and ADL. Arrows in Panel A denote data linkage rather than a causal or chronological flow. The initial recipient is matched from the opposite side of the same forced trade. Forced-side fills are classified jointly by forced-side account, liquidation method, initial recipient, and same-trade ADL role. ADL-selected counterparties are not classified as voluntary private absorbers.

Table A1: Data Sources, Frequencies, Coverage, and Observation Units

Source	Observation unit	Frequency	Coverage	Rows	Main variables
Broad panel: positions	account-asset-window	approximately daily window	215 windows	53,693,108	positions and private position adjustment
Broad panel: accounts	account-window	approximately daily window	215 windows	18,012,686	equity, leverage, portfolio loss, liquidation
Linked portfolios	account-target-asset-window	approximately daily window	215 windows	29,295,083	own and leave-target-out losses, target adjustment
Shift-share risk set	at-risk account-minute	one minute	69,137 accounts; 28 minutes	1,809,043	frozen exposure, oracle innovations, first liquidation
100k policy candidate events	liquidation event	fill/second/minute	216 dates; 196 assets	10,420	position spell, initial order, prices, subsequent flow
Market-liquidation event universe	market-liquidation event	millisecond	216 dates; 329 assets	996,396	account, asset, event time, initial liquidation order
All-route liquidation account-fill times	account-fill time	millisecond	234,267 accounts; market and backstop	1,081,942	account, forced-fill time, market/backstop route indicators
October 10 market clock	asset-minute	one minute	217 assets; 20:00–22:00 UTC	26,257	official spot oracle, mark, mid, impact, spread, depth
Forced-liquidation fills	fill/node	millisecond	October 10 event	338,675	liquidation method, direction, initial recipient, notional
ADL-linked fills	fill/node	millisecond	October 10 event	35,022	forced side, ADL-selected side, notional
Alternative valuation paths	account-minute-scenario	one minute	four stored paths; two in main text	27,924,284	maintenance surplus, shortfall, at-risk status

Notes: All timestamps are UTC.

Rows describe the stored analysis universe, not necessarily the final estimation sample. Table A2 and Table 1 of the main text report the applicable sample restrictions.

The raw data label protocol-directed forced-liquidation fills as user liquidations; they are not user-initiated trades.

Table A2: Broad-Panel Regression Sample Construction

Panel	Sample construction step	Observations	Retained from input (%)	Dropped from prior
A. Private-adjustment sample	Canonical account-asset-window rows	53,693,108	100.000	
A. Private-adjustment sample	Ordinary user-controlled rows in full daily windows	53,570,125	99.771	122,983
A. Private-adjustment sample	Eligible initial cross-margin positions	30,264,840	56.366	23,305,285
A. Private-adjustment sample	Matched endpoint benchmark prices and observed within-window leverage rank	29,488,255	54.920	776,585
A. Private-adjustment sample	Complete path and persistent private-reduction outcome	29,488,255	54.920	0
B. Forced-liquidation sample	Canonical account-window rows	18,012,686	100.000	
B. Forced-liquidation sample	Eligible ordinary user-controlled cross-margin account-windows	5,935,054	32.949	12,077,632
B. Forced-liquidation sample	At least 80% of cross notional benchmark matched	5,887,276	32.684	47,778
B. Forced-liquidation sample	Complete loss and forced-liquidation variables	5,887,276	32.684	0
C. Linked-portfolio sample	Eligible target-position input rows	29,295,083	100.000	
C. Linked-portfolio sample	At least one positive non-target source position	29,295,083	100.000	0
C. Linked-portfolio sample	At least 90% of source notional benchmark matched	27,843,280	95.044	1,451,803
C. Linked-portfolio sample	Complete baseline linked-loss variables	27,843,280	95.044	0
C. Linked-portfolio sample	Complete source-composition controls	27,840,651	95.035	2,629
C. Linked-portfolio sample	Target positions with no own-asset loss (separate subsample)	13,986,132	47.742	

Notes: The private-adjustment sample uses account-asset-window positions. Its initial-balance-sheet restriction keeps ordinary user-controlled accounts with at least \$100 of equity, positive cross-margin notional, finite positive cross-margin leverage, and a positive target position. The final rows also require matched endpoint benchmark prices, an observed within-window leverage rank, valid direction accounting and path reconstruction, and a persistent private-reduction share between zero and one.

The forced-liquidation sample uses account-window observations. Eligibility applies the same ordinary user-controlled-account, \$100 equity, positive cross-notional, and finite positive-leverage conditions. Final rows additionally require benchmark coverage for at least 80 percent of initial cross notional and complete loss, leverage, and forced-liquidation variables.

The linked-portfolio sample uses account-target-asset-window observations. Final rows require at least one other initial position and benchmark coverage for at least 90 percent of non-target source notional. The no-own-asset-loss row is a separate restricted subsample, not the next step in the sequential attrition calculation.

Within each panel, Retained from input uses the first row as its denominator, whereas Dropped from prior compares adjacent sequential rows. The separate no-own-asset-loss subsample has no prior-row attrition entry.

Final dimensions are 190,855 accounts, 195 assets, and 215 windows for the private-adjustment sample; 190,490 accounts and 215 windows for the forced-liquidation sample; and 112,748 accounts, 195 target assets, and 215 windows for the linked-portfolio sample.

B Broad-Panel Margin Fragility and Adjustment

This appendix supports Figure 2 and Table 2 of the main text. The broad panel spans 215 approximately daily windows and documents the distribution of fragility, private adjustment, severe forced liquidation, and portfolio-wide adjustment. The within-window high-leverage rank supplies a comparable measure across daily windows, while the event replay provides the reconstructed distance to the maintenance boundary used in the high-frequency anatomy.

Table B1 reports the full coefficient vectors for persistent private reduction and forced liquidation, including account-fixed-effect specifications. Table B2 contains the targeted event and account checks. Table B3 documents how losses elsewhere in the same collateral pool predict adjustment of a target position, including no-own-loss, account-fixed-effect, and controlled specifications.

Table B1: Full Private-Adjustment and Forced-Liquidation Regressions

Panel	Term	Baseline	Account FE
A. Persistent private reduction	HighLev	3.737*** (0.301)	0.379*** (0.126)
A. Persistent private reduction	Losing leg	0.010 (0.084)	0.004 (0.080)
A. Persistent private reduction	HighLev x losing leg	0.608*** (0.168)	0.219* (0.127)
A. Persistent private reduction	Log initial notional	0.898*** (0.042)	1.646*** (0.079)
B. Forced liquidation	Oracle-implied portfolio loss, per 10 pp	16.205*** (1.483)	19.332*** (1.696)
B. Forced liquidation	HighLev x portfolio loss, per 10 pp	50.614*** (5.229)	47.464*** (4.560)
B. Forced liquidation	Initial leverage x portfolio loss, per 10 pp	-0.086*** (0.012)	-0.087*** (0.012)
B. Forced liquidation	Initial leverage, per 1x	13.672*** (2.088)	13.286*** (3.267)
B. Forced liquidation	Log initial equity, per log point	-0.932** (0.400)	-10.030** (3.945)

Notes: Panel A uses the private-adjustment sample of cross-margin positions: initial account equity of at least \$100, positive cross notional, and finite positive cross leverage. HighLev identifies accounts in the top two within-window leverage deciles. Panel A coefficients are percentage points. Panel B coefficients are percentage-point changes in forced-liquidation notional scaled by initial equity. Loss coefficients are evaluated for a ten-percentage-point oracle-implied portfolio loss; the initial-leverage and log-equity coefficients are evaluated for a one-unit change in their stated units. Both Panel B columns control for continuous initial leverage and its interaction with oracle-implied portfolio loss. Standard errors are two-way clustered by account and asset-window in Panel A and by account and window in Panel B.

Table B2: Broad-Panel Robustness

Panel	Specification	Outcome	Estimate (SE)	Observations	Fixed effects
A. Persistent private reduction	Baseline persistent private reduction	Persistent private reduction	0.608*** (0.168)	29,488,255	asset-window
A. Persistent private reduction	Exclude October 10	Persistent private reduction	0.632*** (0.168)	29,336,324	asset-window
A. Persistent private reduction	Exclude October 9–11	Persistent private reduction	0.682*** (0.169)	29,030,025	asset-window
A. Persistent private reduction	Exclude top 1% active accounts	Persistent private reduction	0.464*** (0.172)	27,631,251	asset-window
A. Persistent private reduction	Endpoint reduction	Endpoint reduction	2.787*** (0.197)	29,488,255	asset-window
B. Forced liquidation	Baseline severity	Forced-liquidation severity	50.614*** (5.229)	5,887,276	window
B. Forced liquidation	Exclude October 10	Forced-liquidation severity	50.685*** (5.293)	5,853,995	window
B. Forced liquidation	Exclude October 9–11	Forced-liquidation severity	53.353*** (5.690)	5,787,309	window

Notes: Panel A uses the private-adjustment sample of cross-margin positions with initial account equity of at least \$100, positive cross notional, and finite positive cross leverage. Panels A and B report the same focal interaction as the corresponding headline specification under the stated sample or outcome change. Private-adjustment estimates are percentage points. Forced-liquidation estimates are percentage-point changes in forced-liquidation notional scaled by initial equity for a ten-percentage-point oracle-implied portfolio loss.

Table B3: Within-Account Linked-Portfolio Adjustment

Panel	Specification	Outcome	Term	Estimate (SE)	Observations	Fixed effects
A. Linked-loss association	Baseline	Persistent private reduction	Linked portfolio loss	14.351*** (1.781)	27,843,280	TA x window
A. Linked-loss association	Controls	Persistent private reduction	Linked portfolio loss	13.807*** (1.693)	27,840,651	TA x window
A. Linked-loss association	No-own-loss target	Persistent private reduction	Linked portfolio loss	19.502*** (5.270)	13,986,132	TA x window
A. Linked-loss association	Account FE	Persistent private reduction	Linked portfolio loss	10.441*** (1.292)	27,840,651	Account + TA x window

Notes: TA denotes target asset. HighLev identifies accounts in the top two within-window leverage deciles and enters each specification as a control. Estimates on linked portfolio loss report outcome percentage points for a one-unit change in the loss-to-equity ratio; a ten-percentage-point change therefore corresponds to one-tenth of the reported estimate. Standard errors are two-way clustered by account and TA x window. The estimates characterize adjustment across positions within the same account-level collateral pool.

C Shift-Share Design for Initial Forced Inventory

This appendix documents the construction and targeted diagnostics for the shock-exposure design in Figure 3 and Table 3. Positions and account risk states are frozen at 20:44 UTC. From 20:45 through 21:12, the design combines those predetermined positions with asset-specific oracle innovations observed at the beginning of each minute. Outcomes are measured during the ensuing minute, and an account leaves the risk set after its first eligible forced liquidation.

C.1 Predetermined Exposure and Relative Oracle Innovations

The pre-event oracle history is sampled on a five-minute grid from August 11 through the 20:44 portfolio freeze on October 10. Let $r_{a\tau}^{O,5m}$ denote the five-minute log change in asset a 's official spot-oracle price. For each eligible asset, the oracle-market factor excludes that asset and weights the remaining returns by lagged dollar open interest:

$$f_{-a,\tau}^{5m} = \frac{\sum_{b \neq a} OI_{b,\tau-1}^{USD} r_{b\tau}^{O,5m}}{\sum_{b \neq a} OI_{b,\tau-1}^{USD}}. \quad (C.1)$$

We estimate the pre-event relation

$$r_{a\tau}^{O,5m} = \alpha_a^{5m} + \beta_a f_{-a,\tau}^{5m} + u_{a\tau} \quad (C.2)$$

separately by asset and freeze the resulting coefficient before the event. Eligible assets must have at least 30 calendar days of history and at least 90 percent coverage over that minimum history.

During the event window, the leave-one-out factor is reconstructed at one-minute frequency with lagged dollar open interest. The source innovation observed at the beginning of minute t is

$$\varepsilon_{at} = r_{at}^{O,1m} - \left(\frac{\alpha_a^{5m}}{5} + \beta_a f_{-a,t}^{1m} \right). \quad (C.3)$$

The frozen-position universe contains 171 source assets with complete event-window innovations and the required pre-event history. An account enters the design when those eligible assets cover at least 90 percent of its frozen cross-margin gross notional.

Let V_{ia0} be account i 's signed dollar exposure to source asset a , and let E_{i0} be its equity at 20:44. The predetermined exposure weight is $s_{ia0} = V_{ia0}/E_{i0}$. It is a leverage-scaled signed exposure and therefore need not lie between zero and one or sum to one across assets. The current external portfolio loss is

$$X_{it} = - \sum_a s_{ia0} \varepsilon_{at}, \quad (\text{C.4})$$

where a positive value is a loss relative to frozen equity. The regression separately controls for the relative loss accumulated through $t - 1$, the current and lagged portfolio losses generated by the common oracle-market component, predetermined balance-sheet controls, minute fixed effects, and frozen risk-state-by-minute fixed effects. Conditional source-sign inference treats the 171 source-asset innovation paths as the primitive shock variation and applies 9,999 sign randomizations to their score contributions.

C.2 Source Concentration and Targeted Robustness

Table C1 reports the effective source counts and concentration statistics separately for liquidation entry and the entry-minute forced-inventory rate.

Table C1: Shift-Share Shock and Exposure Diagnostics

Diagnostic	Liquidation entry	Forced-inventory rate
Nominal source assets	171	171
Effective source count	7.055	6.660
Largest source share (%)	26.661	27.669
Top-five source share (%)	73.187	73.710
Minimum frozen-exposure coverage (%)	90.000	90.000

Notes: Source shares are absolute contributions to the source-level score for the benchmark coefficient. The effective source count is the reciprocal Herfindahl index of those shares.

Eligible source assets must have complete event-window innovations and the required pre-event oracle history. Accounts enter when eligible assets cover at least 90 percent of frozen cross-margin gross notional.

Table C2 contains the diagnostics that bear directly on source dominance and the external origin of the innovation. The leave-source specifications recompute the treatment and its lagged state after removing each account’s largest one or five source exposures. The prior-flow specifications partition each source-minute innovation according to whether same-asset market-method forced liquidation occurred strictly before the oracle snapshot; the retained and excluded components enter the regression jointly.

Table C2: Targeted Robustness and Source-Level Inference

Panel	Specification	Effect (10 pp loss)	Account-clustered SE	Sign p	Path-perm. p	Effective sources	Retained innovation (%)
A. Entry probability	Benchmark	2.021	0.072	0.0001	0.1050	7.055	100.0
A. Entry probability	Leave largest source	1.959	0.092				
A. Entry probability	Leave top five sources	2.708	0.144				
A. Entry probability	No prior market-method flow (1 min)	0.995	0.120	0.4922		6.585	67.5
A. Entry probability	No prior market-method flow (2 min)	2.654	0.146	0.0732		6.741	54.0
B. Forced-inventory rate	Benchmark	1.717	0.065	0.0001	0.1500	6.660	100.0
B. Forced-inventory rate	Leave largest source	1.633	0.083				
B. Forced-inventory rate	Leave top five sources	2.230	0.127				
B. Forced-inventory rate	No prior market-method flow (1 min)	0.781	0.110	0.4974		6.293	67.5
B. Forced-inventory rate	No prior market-method flow (2 min)	2.282	0.132	0.0856		6.558	54.0

Notes: Effects and account-clustered standard errors are percentage-point changes in the stated outcome for a ten-percentage-point external portfolio loss relative to frozen equity. Panel A uses first-liquidation entry; Panel B uses bounded entry-minute forced inventory divided by frozen cross-margin gross notional. Sign p-values use 9,999 conditional sign randomizations of the source-score contributions. Benchmark path-permutation p-values use 1,999 complete-path reassignments within pre-event beta-by-open-interest terciles. Leave-source rows recompute the current innovation and lagged relative-loss state after removing each account’s largest one or five source exposures; source-level p-values are not reported for these account-specific transformations.

Prior-flow rows retain source-minute innovations only when same-asset market-method forced liquidation is absent strictly before the oracle snapshot over the stated interval. The retained and excluded innovation components enter jointly. Effective sources are reciprocal-Herfindahl counts based on absolute source-score contributions.

For the benchmark coefficients, conditional sign randomization rejects the null for both outcomes, whereas complete-path permutation yields p -values of 0.105 and 0.150. Sign randomization preserves each asset’s realized path and identity; permutation reassigns paths within beta-by-open-interest terciles and additionally requires within-stratum exchangeability. Neither procedure is assumption-free; we use the former as the benchmark and the latter as a complementary sensitivity exercise.

The one- and two-minute masks are nested and retain 67.5% and 54.0% of absolute innovation variation, respectively. Their non-monotonic estimates indicate that these restrictions change the composition of source-minute variation; we therefore treat them as sensitivity exercises rather than preferred specifications.

D The \$100,000 Partial-Liquidation Discontinuity

This appendix supports Figure 4 and Panel A of Table 4 in the main text. The running variable is pre-liquidation position value. For long positions entering market-method liquidation near \$100,000, the institutional rule generates a large discontinuity in the initial-order executed share. Because the cutoff changes both the initial-order target and the subsequent 30-second liquidation-handling path, the empirical object is the local reduced-form effect of assignment to the partial-liquidation regime on execution and immediate directional transaction-price stress.

D.1 RD Validity and Alternative-Sample Diagnostics

Figure D1 reports the running-variable density and predetermined balance in the primary long-position sample. Table D1 provides the corresponding formal estimates and a compact diagnostic for the pooled long-and-short sample. Long and short liquidations create opposite directional order flow, and direction composition changes at the cutoff in the pooled data; the primary design therefore estimates the forced-sale effect among long positions.

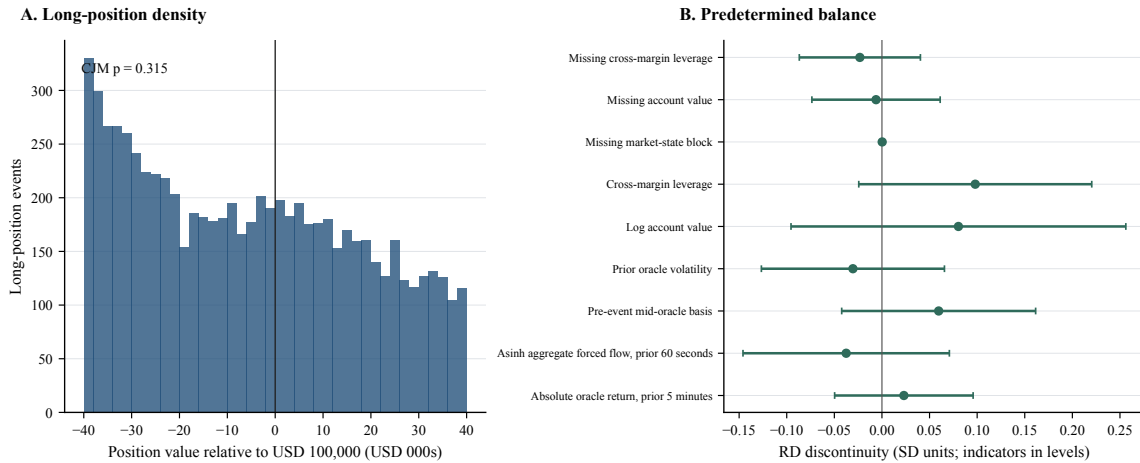


Figure D1: RD Density and Predetermined Balance

Notes: Panel A displays the running-variable distribution for long-position events within \$40,000 of the cutoff; Table D1 reports the formal local-polynomial density test. Panel B reports predetermined balance in the primary long-position sample within \$20,000. Continuous covariates are in standard-deviation units and missingness indicators remain in levels. Error bars denote 95% confidence intervals.

Table D1: RD Validity and Alternative-Sample Diagnostics

Panel	Item	Estimate	Std. error	p-value	N	Notes
A. Primary-sample density	Long positions	-1.85e-06	1.84e-06	0.315	7,368	Cattaneo-Jansson-Ma density difference
B. Balance	Absolute oracle return, prior 5 minutes	0.023	0.037	0.535	3,026	standardized
B. Balance	Asinh aggregate forced flow, prior 60 seconds	-0.038	0.055	0.497	3,562	standardized
B. Balance	Pre-event mid-oracle basis	0.060	0.052	0.252	3,027	standardized
B. Balance	Prior oracle volatility	-0.030	0.049	0.534	3,027	standardized; max one-way clustered SE
B. Balance	Log account value	0.080	0.090	0.371	2,390	standardized
B. Balance	Cross-margin leverage	0.098	0.062	0.117	2,441	standardized
B. Balance	Missing market-state block	0.000247	0.000118	0.037	3,562	indicator
B. Balance	Missing account value	-0.006	0.034	0.857	3,562	indicator
B. Balance	Missing cross-margin leverage	-0.023	0.032	0.475	3,562	indicator
C. Alternative-sample diagnostics	Pooled density difference	3.48e-06	1.51e-06	0.021	10,420	Pooled long-and-short sample
C. Alternative-sample diagnostics	Short-position share discontinuity (pp)	8.184	2.040	6.12e-05	5,107	Pooled long-and-short sample

Notes: Panel A reports the Cattaneo-Jansson-Ma local-polynomial density test for long-position events within USD 40,000 of the cutoff, using a triangular kernel, mass-point adjustment, and data-driven local bandwidths.

Panel B uses the primary long-position sample within USD 20,000. Continuous predetermined covariates are standardized within that bandwidth; missingness indicators remain in levels. Standard errors are two-way clustered by asset and UTC date for market variables and by account and UTC date for account variables. When a two-way covariance is non-positive, the reported standard error is the larger of the corresponding one-way clustered standard errors.

Panel C reports the pooled density test and the discontinuity in direction composition. Long and short liquidations create opposite directional order flow, and the pooled composition changes at the cutoff; the primary design therefore estimates forced-sale effects among long positions. Observations vary across Panel B rows with pre-event data availability.

Table D2 reports the main bandwidth together with narrower and wider bandwidths, a donut sample, exclusion of Bitcoin, complete pre-event market-state coverage, and placebo cutoffs at \$90,000 and \$110,000.

Table D2: Robustness of the Partial-Liquidation RD

Specification	Initial-order executed share (pp)	Execution slippage (bps)	Price stress, 10 sec. (bps)	Max. adverse, 0–10 sec. (bps)	N range
Bandwidth USD 10,000	-50.925*** (2.593)	-17.889** (7.344)	-16.684** (7.097)	-24.824*** (8.362)	1,817–1,857
Bandwidth USD 20,000	-49.998*** (1.964)	-15.571*** (5.420)	-8.267* (4.642)	-15.350** (6.292)	3,490–3,562
Bandwidth USD 40,000	-53.542*** (1.371)	-10.589** (4.718)	-4.159 (3.307)	-12.558*** (4.853)	7,220–7,368
Donut USD 1,000	-51.510*** (2.193)	-13.003*** (5.020)	-9.818** (4.414)	-15.011** (6.402)	3,285–3,350
Exclude BTC	-43.203*** (2.597)	-18.827** (7.576)	-9.413 (9.906)	-15.831 (12.800)	2,025–2,097
Complete market state	-51.102*** (1.923)	-16.223** (6.453)	-8.522* (4.805)	-12.025* (6.514)	2,994–3,026
Placebo USD 90,000	-0.925 (3.118)	23.975 (18.977)	-8.835 (8.703)	0.276 (4.962)	1,437–1,467
Placebo USD 110,000	0.556 (0.415)	1.561 (11.029)	-4.485 (7.644)	3.100 (5.469)	1,368–1,392

Notes: Cells report discontinuity estimates with clustered standard errors in parentheses. The initial-order executed share is reported in percentage points; execution and transaction-price outcomes are reported in basis points. The N range gives the minimum and maximum outcome-specific observations in each specification.

All specifications use triangular-kernel local-linear regressions with separate slopes and asset, UTC-date, and UTC-hour fixed effects. Initial-order-executed-share and execution-slippage standard errors are two-way clustered by account and UTC date; transaction-price standard errors are two-way clustered by asset and UTC date.

The complete-state specification requires all four pre-event market-state inputs. Placebo windows are confined to one side of the true USD 100,000 threshold and therefore do not inherit its institutional first stage. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

D.2 Dynamic and State-Dependent Price Effects

Figure D2 reports the local reduced-form transaction-price effect from zero through 60 seconds after liquidation. The estimates use the primary long-position design at every horizon; the available event sample can vary because a valid transaction price is required at the stated horizon.

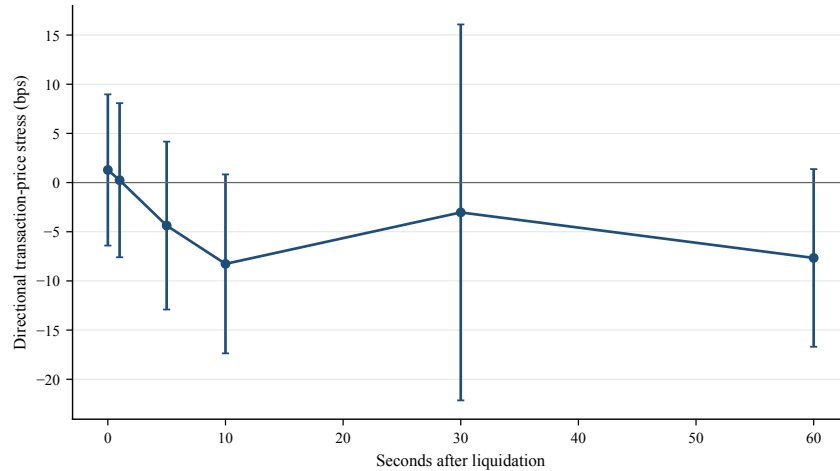


Figure D2: Dynamic Transaction-Price Effects

Notes: The figure reports local reduced-form estimates for the primary long-position sample from zero to 60 seconds after liquidation. Error bars denote 95% confidence intervals. Each horizon uses events with an observed transaction price at that horizon, so sample composition may vary across estimates.

To examine state dependence, we define a pre-liquidation market-stress index as the mean of the standardized absolute five-minute oracle return and standardized aggregate market-method forced-liquidation flow during the preceding 60 seconds. High stress denotes the top quintile of this index in the full $\pm\$40,000$ event panel. Within the main $\pm\$20,000$ bandwidth, the local-linear specification fully interacts the cutoff indicator and local slopes with the high-stress indicator while retaining the main fixed effects and clustering. Table D3 reports the ordinary-state discontinuity, the additional high-stress effect, their sum, and the sensitivity to excluding October 10 while holding the full-sample stress classification fixed.

Table D3: State Dependence and October 10 Sensitivity of the Ten-Second Price Effect

Sample	Ordinary-state discontinuity	High-stress additional	High-stress total	Interaction p-value	N
All dates	-2.530 (4.745)	-25.924** (10.126)	-28.455*** (10.280)	0.011	2,994
Exclude October 10	-5.507 (4.601)	-8.003 (16.321)	-13.511 (16.619)	0.624	2,591

Notes: The dependent variable is directional transaction-price stress ten seconds after liquidation. The high-stress indicator equals one for the top quintile of the pre-liquidation market-stress score, defined as the mean of standardized absolute five-minute oracle returns and standardized aggregate market-method forced-liquidation flow during the preceding 60 seconds.

Each row is a triangular-kernel local-linear interaction regression within USD 20,000 of the policy cutoff. Local slopes are fully interacted with the high-stress indicator; all specifications absorb asset, UTC-date, and UTC-hour fixed effects. Standard errors are two-way clustered by asset and UTC date.

The high-stress total is the sum of the ordinary-state discontinuity and the high-stress additional discontinuity. The exclusion specification retains the original full-sample high-stress cutoff and event labels. The interaction sample requires both complete pre-event stress components and a valid ten-second transaction-price outcome. Stars denote significance at the 1, 5, and 10 percent levels.

D.3 Subsequent-Liquidation Onsets and Feedback

This subsection supports Panel B of Table 4 and Section 5.4 of the main text. It documents the construction of new market-liquidation onsets, the predetermined isolation samples, and the corresponding robustness and placebo estimates.

We begin by sorting all market- and backstop-liquidation fills within account and collapsing records observed for the same account at the same time. This produces 1,081,942 unique account–fill times across 234,267 accounts. An account–fill time begins a new liquidation burst when it occurs more than 30 seconds after that account’s previous forced fill through either route; 1,037,263 observations meet this definition. All 996,396 market-liquidation events match exactly to the all-route account timeline, for a 100% match rate. The resulting onset file contains 986,979 market-onset rows for 231,133 accounts. The pre-onset inactivity test therefore uses all forced-liquidation routes, while an onset itself requires an observed ordinary market liquidation. The outcome measures other accounts newly entering *market* liquidation, not entry through every forced-liquidation route.

For each focal event and horizon, we measure the initial-order notional generated by new same-asset onsets, the number of distinct other accounts generating those onsets, and an indicator for whether any such onset occurs. The focal account is excluded, and each other account is counted once at its first onset in the window. The binary outcome is reported for completeness but is not a primary outcome: its post-event mean ranges from 0.74 to 0.87, leaving little variation on the extensive margin. Within each asset and bandwidth, focal events are sorted by time and retained only when no other eligible focal event occurred during the preceding h seconds. This restriction uses only pre-event information and leaves no overlapping retained forward windows.

Table D4: Predetermined-Isolation Sample Funnel

Sample	Events	Below	Above	Accounts	Assets	Dates	Overlap
Original long RD	3,562	1,811	1,751	2,813	141	200	–
30-second isolation	2,636	1,342	1,294	2,103	141	200	0
60-second isolation	2,492	1,263	1,229	1,986	141	200	0
300-second isolation	2,067	1,046	1,021	1,673	141	200	0

Notes: The original row is the long-position RD sample within USD 20,000 of the policy cutoff. Each isolation row additionally requires complete support for the stated horizon and no other eligible focal event in the same asset and bandwidth during the preceding horizon.

Below and Above classify focal events by pre-liquidation position value. Overlap counts retained forward outcome windows that overlap within asset; the predetermined isolation rule reduces this count to zero.

The Cattaneo–Jansson–Ma density-test p -values are 0.797 in the original long sample and 0.882, 0.777, and 0.701 in the 30-, 60-, and 300-second isolation samples, respectively. These tests all use the $\pm\$20,000$ bandwidth. Table D1 reports 0.315 because its formal density test instead uses the $\pm\$40,000$ window with 7,368 observations; the two values therefore refer to different density-test samples.

The horizon-specific estimates apply a separate isolation sample at 30, 60, and 300 seconds. Their coefficients cannot be read horizontally as a dynamic path because both the outcome horizon and the focal-event sample change. Panel B of Table 4 fixes all three horizons to the strict 300-second sample of 2,067 events. Panel B of Table D5 reports the horizon-specific estimates so that the two output conventions can be reconciled. Panel A reports robustness of the 300-second outcomes. The number of newly liquidated accounts is negative in every specification, whereas the notional estimate is close to zero in the $\pm\$40,000$ bandwidth.

Panel C of Table D5 reports all nine predetermined placebo estimates. Pre-event notional and account-count coefficients are insignificant at every horizon. The binary indicator is discontinuously lower at 30 seconds (-0.0720 , $p = 0.038$) and 60 seconds (-0.0633 , $p = 0.040$). Because this indicator is saturated both before and after the focal event and has no post-event policy effect, it is not used as a primary outcome.

Table D5: Downstream-Liquidation Robustness, Horizon-Specific Estimates, and Placebos

<i>Panel A. Robustness of 300-Second Downstream Outcomes</i>			
Specification	New liquidation notional (\$m)		Newly liquidated accounts
No fixed effects	-0.256***	(0.089)	-17.514** (7.115)
Main specification	-0.357**	(0.140)	-22.386*** (7.745)
Full controls	-0.735***	(0.172)	-34.210*** (7.174)
Bandwidth $\pm$ \$10,000	-0.424	(0.609)	-32.732 (24.577)
Bandwidth $\pm$ \$40,000	0.005	(0.260)	-15.366* (8.863)
Exclude October 10	-0.207	(0.307)	-19.603* (10.620)
Exclude BTC	-0.288	(0.565)	-16.567 (13.978)
Allow overlapping windows	-0.958**	(0.447)	-29.199 (18.073)
Post minus pre	-0.426***	(0.153)	-24.042** (9.527)
<i>Panel B. Horizon-Specific Isolation Samples</i>			
Horizon and observations	New liquidation notional (\$m)		Newly liquidated accounts
30 seconds ($N = 2,636$)	0.038	(0.055)	3.675 (3.409)
60 seconds ($N = 2,492$)	-0.190	(0.155)	-2.587 (2.956)
300 seconds ($N = 2,067$)	-0.357**	(0.140)	-22.386*** (7.745)
<i>Panel C. Predetermined Pre-Event Placebos on the Common Sample</i>			
Outcome	30 seconds	60 seconds	300 seconds
New liquidation notional (\$m)	0.0509 (0.0386); p=0.188	0.0603 (0.0434); p=0.165	0.0693 (0.0424); p=0.102
Newly liquidated accounts	1.295 (1.755); p=0.461	0.396 (2.347); p=0.866	1.656 (2.491); p=0.506
Any new liquidation	-0.0720** (0.0346); p=0.038	-0.0633* (0.0307); p=0.040	-0.0364 (0.0346); p=0.292

Notes: Panel A reports 300-second reduced-form discontinuities. The main specification uses the common 2,067-event sample and absorbs asset, UTC-date, and UTC-hour fixed effects; standard errors are two-way clustered by asset and UTC date. Across specifications, the account-count coefficient is always negative. The notional estimate is close to zero in the plus-or-minus USD 40,000 bandwidth, motivating the emphasis on the breadth of affected accounts.

Panel B reports estimates in which each horizon uses its own predetermined-isolation sample. Because the observations change across columns, these coefficients are not a dynamic path. Panel B of Table 4 instead fixes all horizons to the common 2,067-event sample.

Panel C reports common-sample pre-event outcomes. Cells contain coefficients, clustered standard errors in parentheses, and p-values. The notional and account-count placebos are insignificant at all horizons. The binary indicator is imbalanced at 30 and 60 seconds, but is saturated before and after the focal event and has no post-event policy effect; it is reported for completeness rather than used as a primary outcome. Three, two, and one asterisks denote significance at the 1%, 5%, and 10% levels.

D.4 Robust Bias-Corrected Inference

Table D6 supplements the fixed-bandwidth estimates with the Calonico–Cattaneo–Titiunik robust bias-corrected local-polynomial procedure. The estimates use a local-linear polynomial, local-quadratic bias correction, a triangular kernel, and a common MSE-optimal bandwidth selected separately for each outcome. We report both the canonical unadjusted estimator and a version adjusted for asset, UTC-date, and UTC-hour indicators. The initial-order discontinuity remains essentially unchanged. Execution-cost and local-price estimates retain their direction and are statistically significant in the indicator-adjusted specifications. The subsequent account-count estimates remain negative but are less precise. Subsequent notional is the exception: the indicator-adjusted CCT estimate reverses sign to +\$1.544 million, but its selected bandwidth contains 403 effective observations against 361 retained indicators. We therefore do not interpret this near-saturated specification as a stable local estimate. The unadjusted CCT point estimate is a \$0.023 million reduction with a \$0.706 million standard error, so the CCT exercise provides no robust evidence of a negative notional discontinuity. Accordingly, account count is the primary downstream outcome and notional is complementary.

For scale, the fuzzy-RD translation of the 300-second estimates implies that a 10-percentage-point increase in the initial liquidation share corresponds to approximately \$77,500 of additional new-liquidation notional and 4.86 additional accounts. The design estimates the reduced-form effect on subsequent liquidation and does not separately decompose the price-mediated channel; Section 2 of the main text defines that scope.

Table D6: Robust Bias-Corrected Inference for the \$100,000 Discontinuity

Outcome	Main	CCT robust bias correction		MSE-optimal bandwidth	
	Fixed \$20,000 (1)	Unadjusted (2)	Indicator-adjusted (3)	Unadjusted (4)	Adjusted (5)
<i>Panel A. Initial Execution and Local Price Outcomes</i>					
Initial liquidation-order share	-49.998*** (1.964)	-49.022*** (3.321)	-50.869*** (2.265)	10,013	13,199
Execution slippage	-15.571*** (5.420)	-23.619* (13.367)	-20.462*** (7.420)	8,105	7,980
Transaction-price stress at 10 seconds	-8.267* (4.642)	-20.634* (11.569)	-11.789** (5.621)	9,210	6,024
Maximum adverse move, 0–10 seconds	-15.350** (6.292)	-29.349 (17.969)	-27.477*** (9.271)	9,144	8,970
<i>Panel B. Subsequent Liquidation at 300 Seconds</i>					
New liquidation notional, 300 seconds	-0.357** (0.140)	-0.023 (0.706)	1.544*** (0.458)	7,474	3,185
Newly liquidated accounts, 300 seconds	-22.386*** (7.745)	-10.841 (53.258)	-30.394* (17.263)	5,750	4,424

Notes: Column (1) reproduces the fixed-\$20,000-bandwidth estimates from Table 4, with asset, UTC-date, and UTC-hour fixed effects and the two-way clustering described there. Columns (2)–(5) report Calonico–Cattaneo–Titiunik robust bias-corrected local-linear estimates with local-quadratic bias correction, a triangular kernel, and MSE-optimal common bandwidths. Column (2) is unadjusted; column (3) includes asset, UTC-date, and UTC-hour indicators. The supplementary CCT estimates use CR1 standard errors clustered by account for initial-order share and slippage and by asset for price and subsequent-liquidation outcomes; the software does not reproduce the main table’s two-way covariance estimator. The indicator-adjusted feedback specifications retain 361 indicator columns and use 403 effective observations for notional and 539 for account count, so they are read jointly with the unadjusted estimates. Robust standard errors are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

E October 10 Phase-Aligned Adjustment

This appendix places private position reduction by high-fragility accounts and forced liquidation on the event phases used in Figure 5 of the main text. High-fragility accounts are those in the top two minute-specific fragility deciles in the event replay. Table E1 reports phase totals for the 29-minute lead-in and eight-minute core burst on the same minute-level clock.

Table E1: Phase-Aligned Private and Forced Adjustment

Adjustment	Lead-in (USD bn)	Core burst (USD bn)
High-fragility private position reduction	0.675	0.439
Forced liquidation	0.602	5.643

Notes: The lead-in covers 20:44–21:12 UTC (29 minutes), and the core burst covers 21:13–21:20 UTC (eight minutes). Reported amounts are phase totals rather than per-minute rates.

High-fragility accounts are those in the top two minute-specific fragility deciles in the event replay. Private position reduction sums their voluntary reducing flow; forced liquidation is measured on the same minute-level event clock. Amounts are USD billions.

F Stock–Flow Reconciliation for the October 10 Event

This appendix supports Figure 6 and Panels A–B of Table 5 in the main text. The account-state paths are observed at one-minute replay checkpoints, whereas forced liquidation is measured from transaction-level fills. Panels A and B of Figure 6 retain, within each valuation scenario, one observation per account at its first observed margin shortfall during the core burst. They therefore measure first-observed balance-sheet stocks. Transaction-level fills record forced flow throughout the event, including boundary crossings that occur between replay checkpoints.

The first-observed convention includes accounts already observed in shortfall during the lead-in. Of the 1,940 accounts under the realized mark path and 736 accounts under the official spot-oracle benchmark reported in Figure 6, 57 and 56, respectively, were already in shortfall during the lead-in. The two scenario sets are not nested: 621 accounts enter the core-burst shortfall set under both paths, 1,319 only under the realized mark path, and 115 only under the spot-oracle benchmark. The \$161.9 million headline difference is the net difference between the two scenario-level stocks.

The event-wide gross forced-transfer measure cumulates all forced-side market- and backstop-method fills across accounts and time. It contains 338,675 liquidation rows for 34,042 accounts and totals \$6.666 billion. Figure 6 instead measures \$657.4 million of gross position notional once for 1,940 accounts at their first observed core-burst shortfall. Panel C of Figure 6 provides the comparable flow object by cumulating subsequent realized liquidation fills for the predefined first- and second-wave accounts.

G Short-Horizon Markouts on Initial HLP/System Receipt

The main text traces the October 10 risk-transfer chain from accounts outside the identified HLP/system set to public-market counterparties, HLP/system accounts, and ADL-selected counterparties. This appendix examines the short-horizon price exposure borne by identified HLP/system accounts when they receive non-ADL liquidation fills. We identify the recipient from the opposite side of the same matched trade and retain recipients in the versioned HLP/system registry.

An on-chain forensic analysis by Jia et al. (2026) follows the vault’s full liquidation-to-unwind pipeline using a six-hour benchmark and reports positive longer-horizon backstop markouts and approximately \$50 million in net vault profit. Our measure stops at one, five, and thirty minutes after initial receipt and excludes subsequent trading, hedging, funding, and position unwinding; the positive thirty-minute markout is therefore qualitatively aligned with their longer-horizon finding but is neither the same estimand nor a net-P&L calculation.

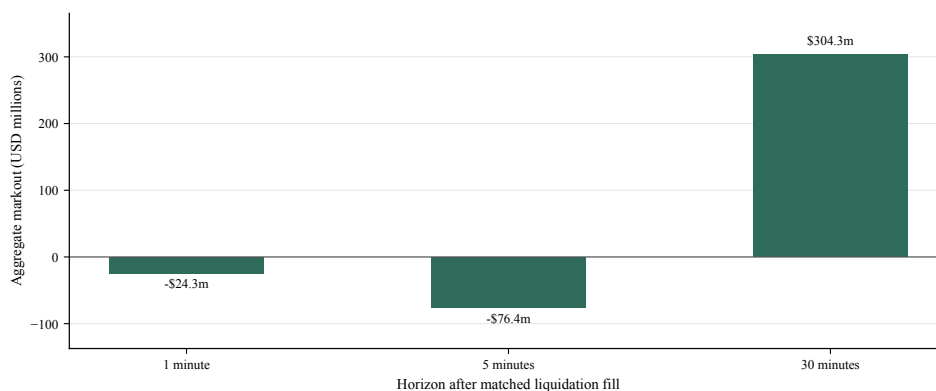


Figure G1: Short-Horizon Markouts on Identified HLP/System Receipt

Notes: The bars sum signed dollar markouts across 46,927 non-ADL liquidation fills whose matched recipient belongs to the four-account identified HLP/system registry. For each fill, the markout equals received notional times the recipient’s position direction times the percentage change from the fill price to the future Hyperliquid mid price; the mark price and then the oracle price are used only when the mid is unavailable. Price coverage at the one-, five-, and thirty-minute horizons is 100.00%, 99.98%, and 99.00%, respectively. Positive values indicate that the subsequent price movement favors the received position. The measure captures short-horizon exposure on matched fills and excludes subsequent trading, hedging, and funding. It is not oracle-adjusted and therefore combines the local execution price with the subsequent external market path.